The Root-Locus Method of Evans

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1 Introduction to the Root Locus

- Consider a system with the standard loop

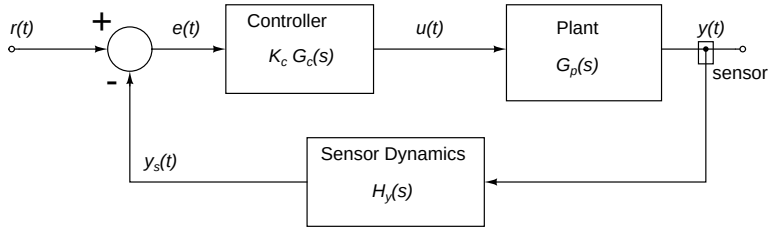


Figure 1: Standard series feedback.

- The compensator is given a parameter K_c, which adjusts the loop gain.
- The closed-loop transfer function is:

$$T_{ry}(s) = \frac{K_c G_c(s) G_p(s)}{1 + K_c G_c(s) G_p(s) H_y(s)} \tag{1}$$

- If we write out the numerator and denominator terms, and clear the fractions within the fraction, $T_{ry}(s)$ is given as:

$$T_{ry}(s) = \frac{K_c \frac{N_c(s) N_p(s)}{D_c(s) D_p(s)}}{1 + K_c \frac{N_c(s) N_p(s) H_y(s)}{D_c(s) D_p(s) D_y(s)}} = \frac{K_c N_c(s) N_p(s) D_y(s)}{D_c(s) D_p(s) D_y(s) + K_c N_c(s) N_p(s) N_y(s)} \tag{2}$$

- The closed-loop poles are given as the roots of

$$D_c(s) D_p(s) D_y(s) + K_c N_c(s) N_p(s) N_y(s) = 0 \tag{3}$$

- As K_c is varied from 0 .. ∞, the closed-loop poles will move smoothly on the S-plane.
- The paths swept out by the closed-loop poles are called the ***root loci***, and reveal important characteristics of controller designs.

1.1 A first root-locus example, determined numerically

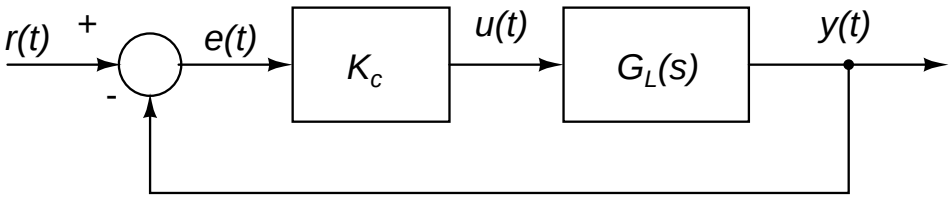


Figure 2: Simple closed-loop system with parameter K_c .

- Starting with an open-loop transfer function

$$G_L(s) = \frac{s+10}{s(s+4)} \tag{4}$$

1. Plot the open-loop pole-zero constellation

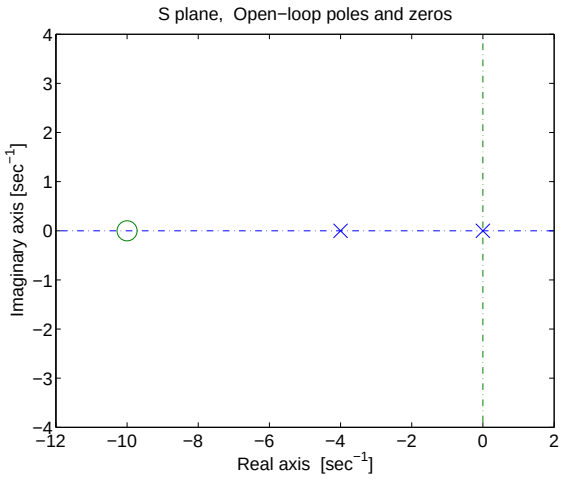
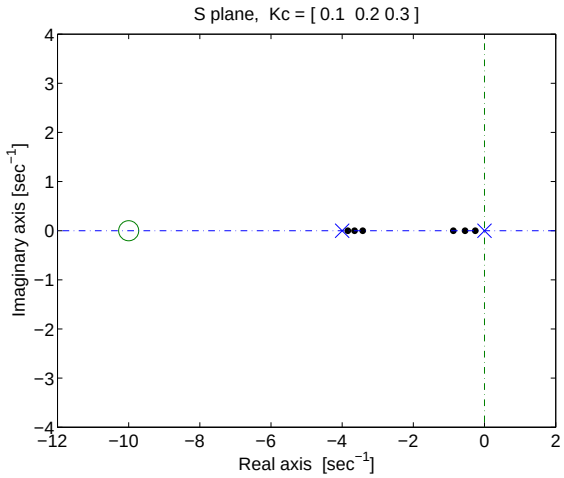


Figure 3: Open-loop pole-zero constellation, starting point for the root locus.

2. Calculate the closed-loop poles as K_c increases from 0 ... 0.3




```
Kcs = [ 0.1 0.2 0.3]
jKc = 0;
for Kc = Kcs,  jKc=jKc+1;
  CLPoles(:, jKc) = roots( Dp+Kc*[0 Np] );
end

plot(real(OLPoles), imag(OLPoles), 'x', ..
   real(OLZeros), imag(OLZeros), 'o', ...
   real(CLPoles), imag(CLPoles), 'k.')
```

Figure 4: A root locus, made by calculating the closed-loop poles for K_c between 0 and 0.3.

3. For the numerical example, keep going, K_c goes 0 ... 10, then 0 ... ∞

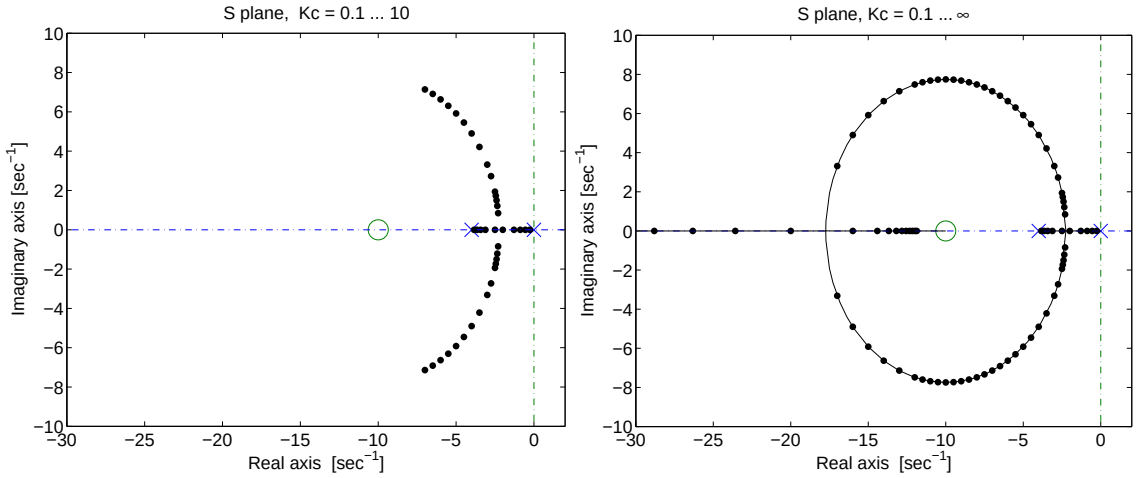


Figure 5: Root-locus branches keep growing as K_c increases. Finally, the full range is covered.

- The root-locus branches tell us where the closed-loop poles can be
  - Farther to right = faster system,
  - Farther vertically = less damped system
  - etc.
- In this example, as K_c is increased:
  - Dominant mode gets faster, then gets slower
  - Dominant mode gets more oscillatory, then gets better damped.

1.2 Root-locus demonstration, example 2

- System under consideration

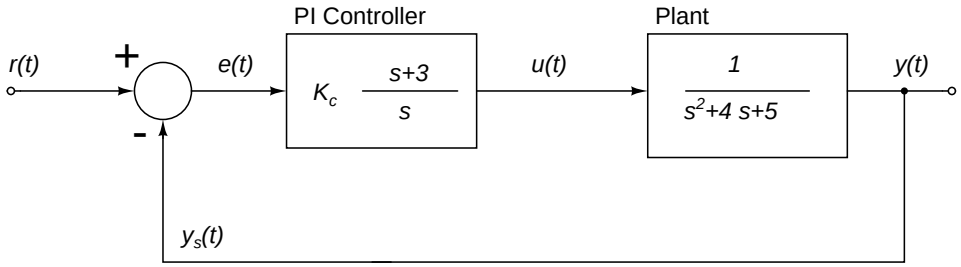
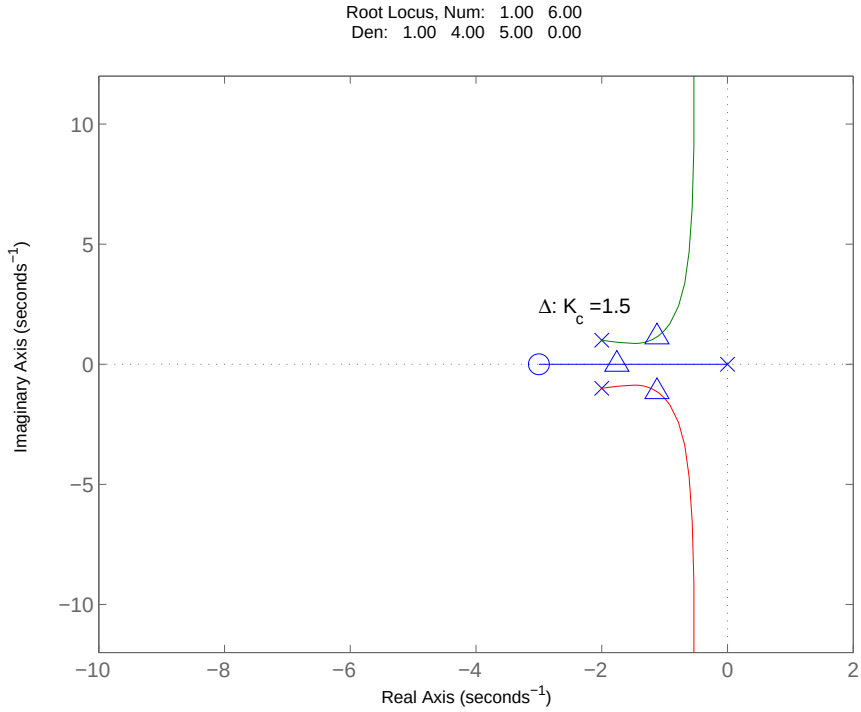


Figure 6: Block diagram of system under consideration.

- Root locus, at each value of K_c there are 3 poles.




```
Gc = tf([1, 3], [1 0])
Gp = tf(1, [1 4 5]);
rlocus(Gc * Gp);

Kc = 1.5;
Try = feedback(Kc*Gc*Gp, 1);
CLPoles = pole(Try);
hold on
 plot(real(Poles), ...
      imag(Poles), ...
      '^');
hold off
```

Figure 7: Root locus example, plant with compensator.

- For this system, we can see:
  - For small K_c , gets faster as K_c increases (real branch moves left).
  - $K_c > 1.5$, system becomes quite oscillatory (complex branches move out).

1.3 Root-locus demonstration, example 3

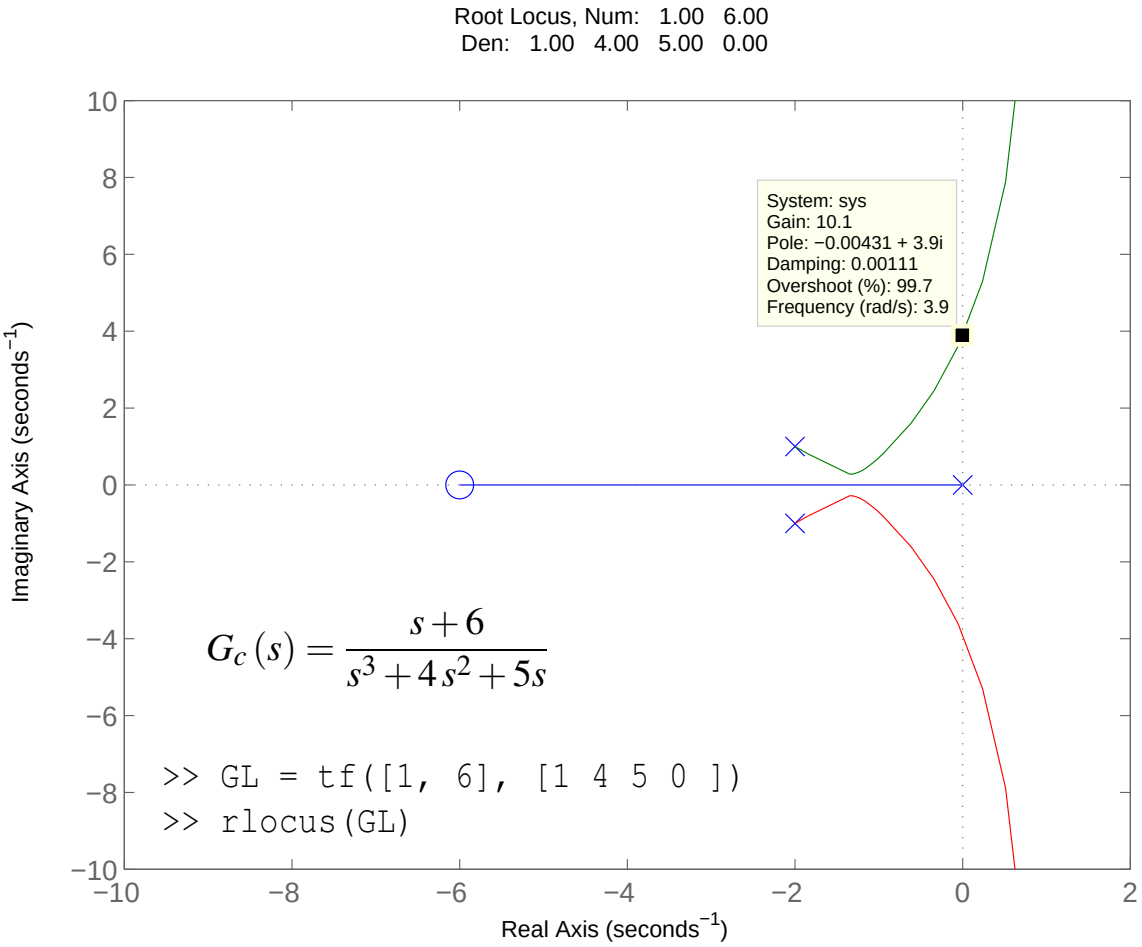


Figure 8: Root locus example.

- Left clicking on root-locus branch opens a dialog box
- This system becomes unstable at $K_c = 10.1$

1.4 Summary, Root-locus and controller design

- Recall that controller design proceeds in two steps:
  1. Determination of the over-all controller structure (PI, PID, phase-lead, etc.)
  2. Tuning of the specific controller parameters
- The root-locus method is enabling for both steps.

2 Mathematical foundations of the root locus

- We are interested in the characteristic equation of a closed-loop system, such as seen in figure 9.

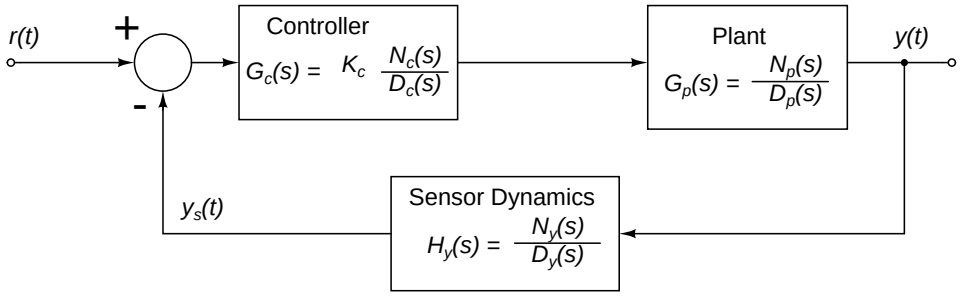


Figure 9: Generic single-loop block diagram.

- The transfer function is given by:

$$Try(s) = \frac{Y(s)}{R(s)} = \frac{K_c G_c(s)\ G_p(s)}{1 + K_c G_c(s)\ G_p(s)\ H_y(s)} \tag{5}$$

- Writing the characteristic equation, pulling out K_{rlg}

$$\begin{aligned} 1 + K_c G_c(s)\ G_p(s)\ H_y(s) &= 1 + \frac{K_c N_c(s)\ N_p(s)\ N_y(s)}{D_c(s)\ D_p(s)\ D_y(s)} = 0 \\ &= 1 + K_{rlg} \frac{b(s)}{a(s)} = 0 \end{aligned} \tag{6}$$

where:

$$\frac{K_c N_c(s)\ N_p(s)\ N_y(s)}{D_c(s)\ D_p(s)\ D_y(s)} = K_{rlg} \frac{b(s)}{a(s)} \tag{7}$$

  - G_c, N_c, D_c, etc. are illustrated in figure 9;
  - $b(s)$ is the zeros polynomial, scaled to be monic;
  - $a(s)$ is the poles polynomial, scaled to be monic;
  - K_c is a controller parameter, and K_{rlg} is the root-locus gain.

Basics (continued)

- For this part, define $G_L(s)$ to be "the root-locus loop gain",

$$K_{rlg}\, G_L(s) = K_{rlg}\, \frac{b(s)}{a(s)} \tag{8}$$

  where $b(s)$ and $a(s)$ are monic (any constant comes out into the K_{rlg}).

- For example, for the system seen in figure 10,

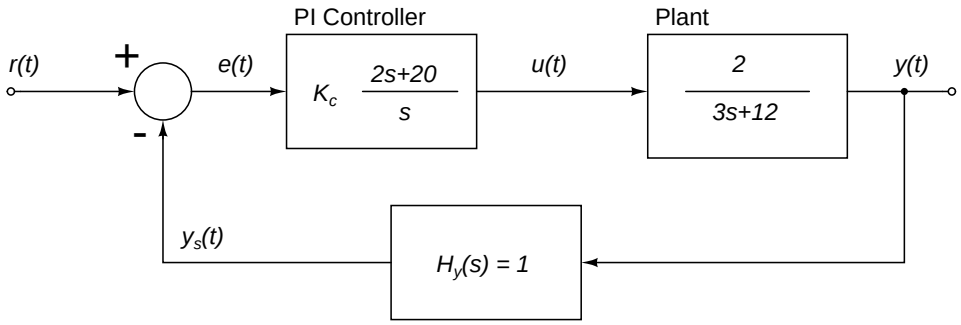


Figure 10: Block diagram of system under consideration.

$$K_c\, G_c(s) = K_c\, \frac{2s+20}{s}, \quad G_p(s) = \frac{2}{3s+12}, \quad H_y(s) = 1$$

Then

$$\begin{aligned} K_{rlg}\, G_L(s) &= K_c\, G_c(s)\, G_p(s)\, H_y(s) = K_c\, \frac{2s+20}{s}\, \frac{2}{3s+12} \\ &= K_c\, \frac{4s+40}{3s\,(s+4)} \end{aligned}$$

which gives

$$K_{rlg} = K_c\, \frac{4}{3} \quad , \quad G_L(s) = \frac{s+10}{s\,(s+4)}$$

Making the $b(s)$ and $a(s)$ monic

$$K_{rlg}\,G_L(s) \;=\; K_{rlg}\,\frac{b(s)}{a(s)} = \frac{4}{3}\,K_c\,\frac{s+10}{s\,(s+4)} \tag{9}$$

- From Eqn (9) we see that

$$K_{rlg} = \frac{4}{3}K_c \quad , \qquad G_L(s) = \frac{s+10}{s\,(s+4)} \tag{10}$$
$$b(s) = s+10$$
$$a(s) = s\,(s+4)$$

- Considering the details of the previous page, the characteristic equation can be expressed in four equivalent ways, each of which plays a role in understanding the root locus:

$$i) \qquad 1+K_{rlg}\,G_L(s) \;=\; 0 \tag{11}$$

$$ii) \qquad 1+K_{rlg}\,\frac{b(s)}{a(s)} \;=\; 0 \tag{12}$$

$$iii) \qquad a(s)+K_{rlg}\,b(s) \;=\; 0 \tag{13}$$

$$iv) \qquad G_L(s) \;=\; -\frac{1}{K_{rlg}}\,, \quad K_{rlg} = \frac{-1}{G_L(s)} \tag{14}$$

  Where:

  - Eqns (i) - (iii) are derived from Eqn (6); and
  - Eqn (iv) derives from Eqn (i).

- Because $b(s)$ and $a(s)$ are monic polynomials, they can be directly determined from the open-loop poles and zeros.

2.1 Foundation of the Root Locus: the Angle and Magnitude Conditions

- Forms *i) .. iv)* for the characteristic equation are given above.
- Consider the form *iv)* for the characteristic equation, from which we find:

$$G_L(s) = -1/K_{rlg} \tag{15}$$

  - Recall that for any value of s, $G_L(s)$ evaluates to a complex number.
    1. $-1/K_{rlg}$, expressed as a complex number in polar coordinates gives:

$$-\frac{1}{K_{rlg}} = \left|\frac{1}{K_{rlg}}\right| \angle 180^o + 360^o(l-1) \tag{16}$$

       where l is an integer.
    2. The angle is $180^o + 360^o(l-1)$ because -1 in polar coordinates is $1\angle 180^o$; and the complex value repeats every 360°:

$$\begin{array}{ccccccc} -1 = ... = 1\angle -180^o & = & 1\angle +180^o = 1\angle +540^o ... \\ ... \quad l=0 & & l=1 & & l=2 & & ... \end{array}$$

- Considering points 1 and 2, for some point s_0 on the S-plane to satisfy Eqn (15), $G_L(s_0)$ must satisfy two conditions:

  Angle Condition: $\angle G_L(s_0) = 180^o + 360^o(l-1)$, where l is an integer.

  - The Angle Condition determines what points are in the root locus:

    Any point s_0 on the S-plane which satisfies the angle condition is a closed-loop pole of the system for $K_{rlg} = \frac{1}{|G_L(s_0)|}$.

    Key to Root Locus

  Magnitude Condition: $K_{rlg} = \frac{1}{|G_L(s_0)|}$.

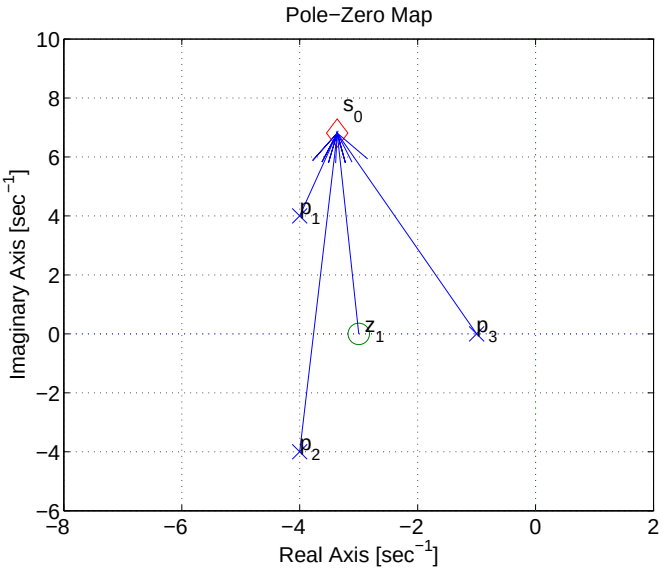
  - The Magnitude Condition can be satisfied for any $G_L(s)$ with $\angle G_L(s) = 180^o \pm (l-1)$ by setting $K_{rlg} = 1/|G_L(s)|$.
  - The magnitude condition determines the value of K_{rlg} to put a pole at point $s = s_0$.

2.1.1 Example of Magnitude and Angle Conditions

- Because the numerator and denominator of $G_L(s)$ are monic, for any value of s_o,

$$G_L(s_0) = \frac{\Pi_{i=1}^{m} \text{ Vectors from Zeros to } s_0}{\Pi_{i=1}^{n} \text{ Vectors from Poles to } s_0}$$

  which is illustrated in figure 11.



$$\begin{aligned} p_1 &= -4 + j4 \\ p_2 &= -4 - j4 \\ p_3 &= -1 \\ z_1 &= -3 \\ s_0 &= -3.37 + j6.81 \end{aligned}$$

Figure 11: $G_L(s)$ give by vectors from poles and zeros.

- In the example of figure 11,

$$\begin{aligned} G_L(s_0) &= \frac{(-0.37 + j6.81)}{(0.63 + j2.81)(0.63 + j10.81)(-2.37 + j6.81)} \\ &= \frac{6.82\angle 93.1^o}{2.88\angle 77.3^o \; 10.83\angle 86.6^o \; 7.21\angle 109.1^o} = 0.30\angle -180.0^o \end{aligned}$$

- Point $s_0 = -3.37 + 6.81\,j$ is in the root locus.

- Point s_0 is a closed-loop pole when $K_{rlg} = |1/G_L(s_0)| = 3.33$.

Example (continued)

- The example of figure 11 is given from:

$$G_L(s) = \frac{s+3}{(s+1)(s^2+8s+32)}$$

- The root locus is seen in figure 12, with s_0 marked.

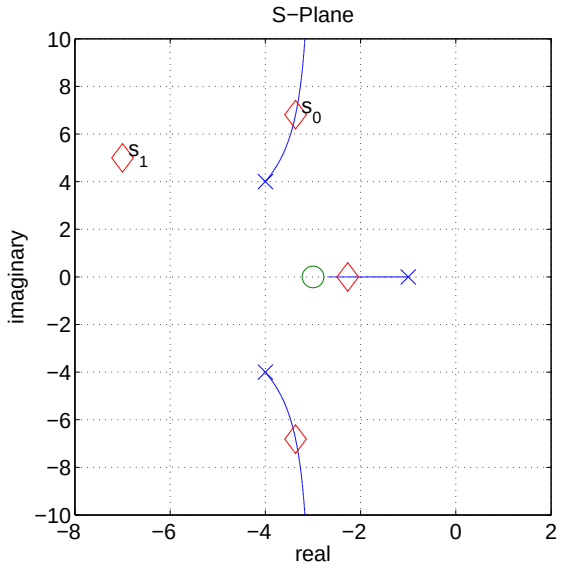
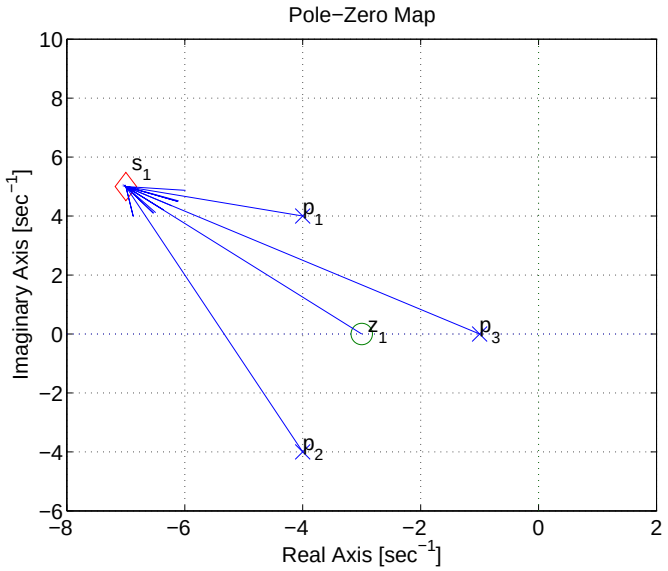


Figure 12: Root locus of example of figure 11.

Example (continued)

- A second point s_1 is seen in figure 12.
- As seen in figure 12, s_1 will violate the Angle Condition, because it is off the root locus. Point s_1 is analyzed in figure 13.



$$p_1 = -4 + j4$$
$$p_2 = -4 - j4$$
$$p_3 = -1$$
$$z_1 = -3$$
$$s_1 = -7 + j5$$

Figure 13: $G_L(s)$ give by vectors from poles and zeros, at point s_1.

- $G_L(s_1)$ is given by:

$$G_L(s_1) = \frac{(-4+j5)}{(-3+j1)(-3+j9)(-6+j5)}$$

$$= \frac{6.40\angle 128.7^o}{3.16\angle 161.6^o \, 9.49\angle 108.4^o \, 7.81\angle 140.2^o} = 0.027\angle -78.5^0$$

- $G_L(s_1)$ satisfies the magnitude condition for $K_{rlg} = 1/0.027$; but
- $G_L(s_1)$ does not satisfy the angle condition;

$\Longrightarrow G_L(s_1)$ is not in the root locus for any value of K_{rlg}.

2.2 Terminology

- Root loci exhibit nine distinct phenomena that are useful for hand sketching and interpreting the locus. The distinct phenomena are:
  1. Root-locus branch . . . . . (the root loci have distinct branches)
  2. Number of branches . . . . . (equals the number of OL poles)
  3. Break out (from the real axis) . . . . . (2 branches, $\pm 90^o$)
  4. Break in (to the real axis) . . . . . (2 branches, $\pm 90^o$)
  5. Departure angle . . . . . (angle of locus departing from a pole)
  6. Arrival angle . . . . . (angle of a locus arriving to a zero)
  7. Asymptotes . . . . . (limit as $K \to \infty$)
  8. Centroid . . . . . (center of asymptotes)
  9. Frequency of crossing the imaginary axis . . . . . (branches go unstable)
- These are defined with the aid of figure 14, which is a sufficiently rich locus to exhibit all of the phenomena.

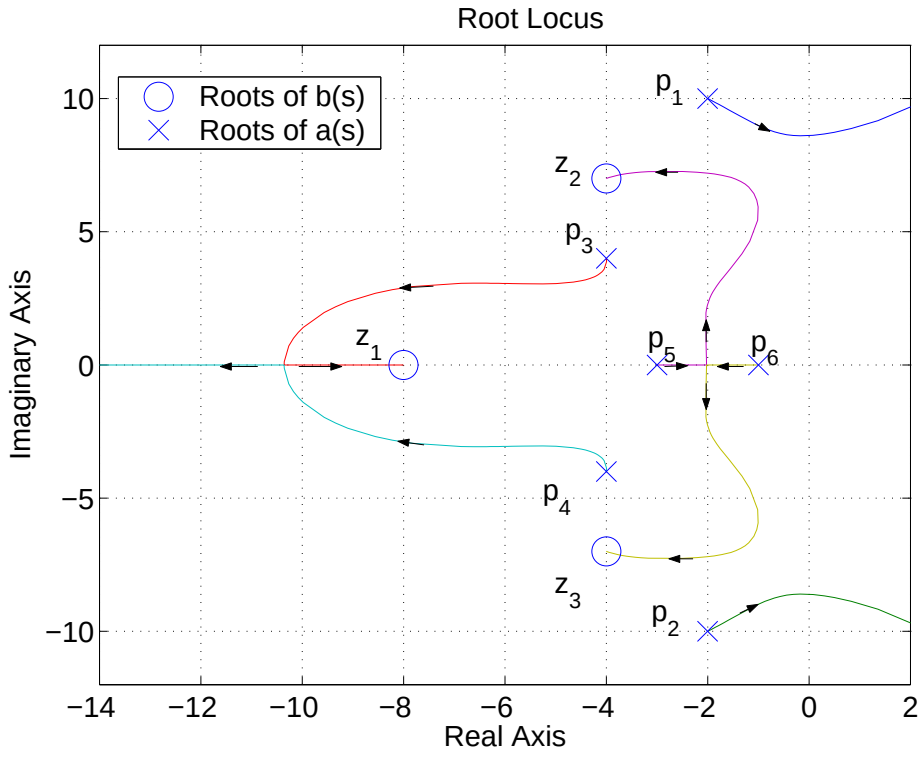


Figure 14: Example root locus showing nine distinct phenomena.

2.2.1 Number of root-locus branches

- A "root-locus branch" is the path one root follows as K_{rlg} goes from 0 to ∞.
- It is seen in figure 15 that a root-locus branch starts at each open loop pole.

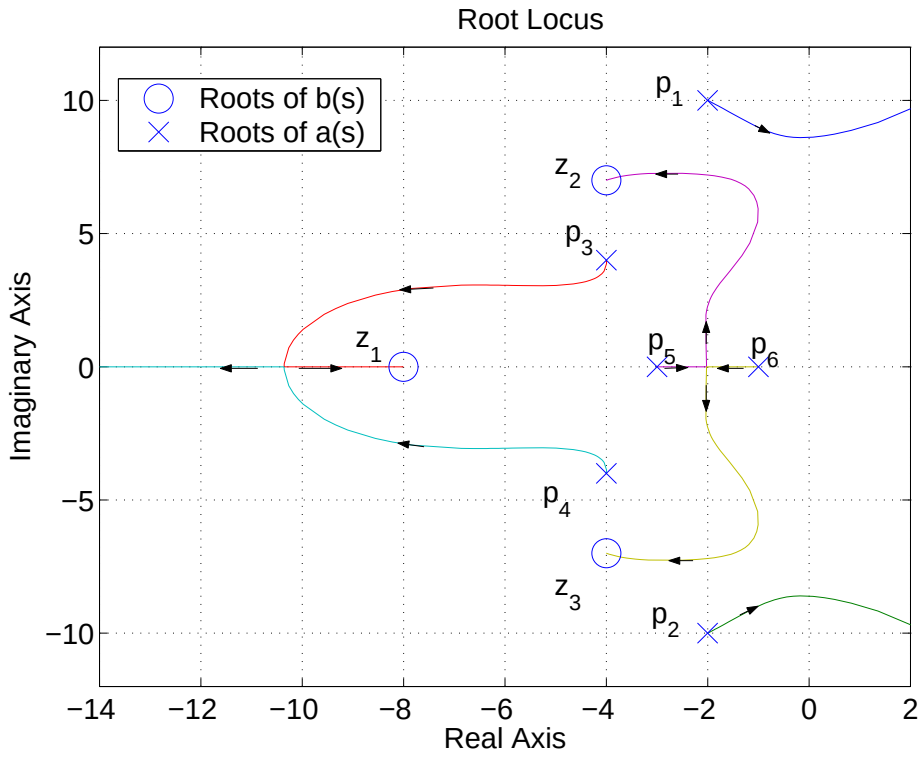


Figure 15: Example root locus showing eight distinct phenomena (repeated).

- The loop gain of the system of figure 15 is given by:

$$K_{rlg}\,G_L(s) \;=\; K_{rlg}\,\frac{(s+8)\,(s+4\pm j7)}{(s+1)\,(s+3)\,(s+4\pm j4)\,(s+2\pm j10)} = K_{rlg}\,\frac{b(s)}{a(s)} \quad (17)$$

- n, the number of poles (order of the $a(s)$ polynomial), is 6.
- m, the number of zeros (order of the $b(s)$ polynomial), is 3.

Number of branches (continued)

- As seen in figure 15, a root-locus branch originates at every open-loop pole.
  - Since the example has 6 open-loop poles, there are 6 root locus branches.

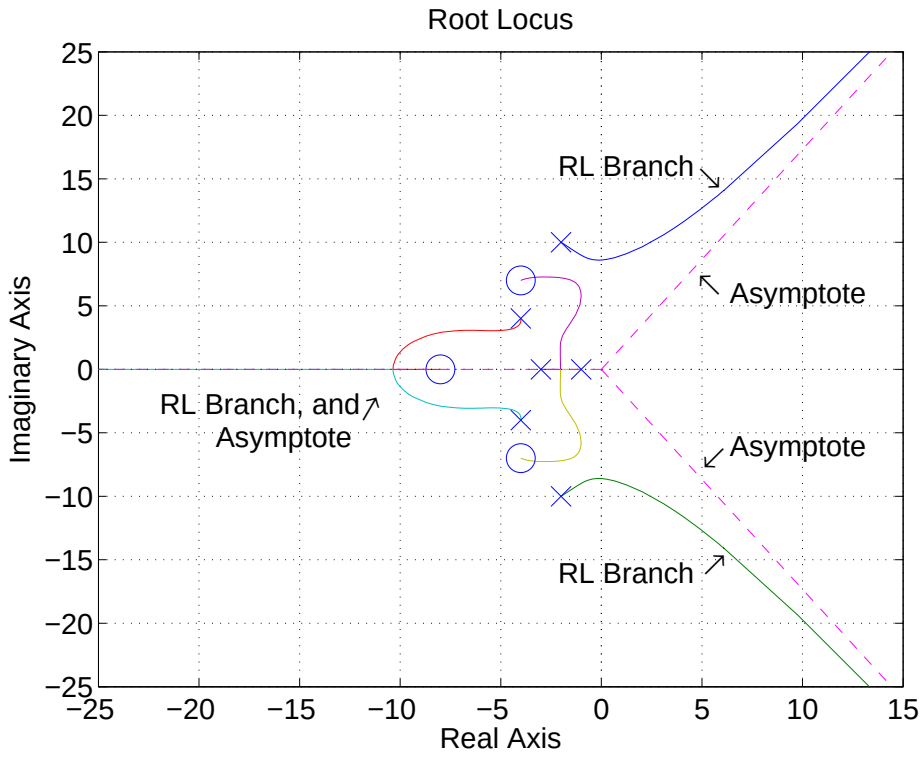


Figure 16: Example root locus showing eight distinct phenomena, expanded view showing asymptotes.

- Figure 16 illustrates the ***asymptotes.***
  - The n root-locus branches start at the n (6) poles
  - m of these branches travel to the m (3) zeros
  - The $n-m$ remaining branches (3) go out to infinity.

    $\rightarrow$ These branches go to infinity along $(n-m)$ asymptotes.

2.2.2 Break out

- When 2 root loci come together on the real axis; they join and then break out into the imaginary plane; as seen in figure 17.
- The initial angle at a break out is $\pm 90^o$, as seen in the blow up (figure 18).

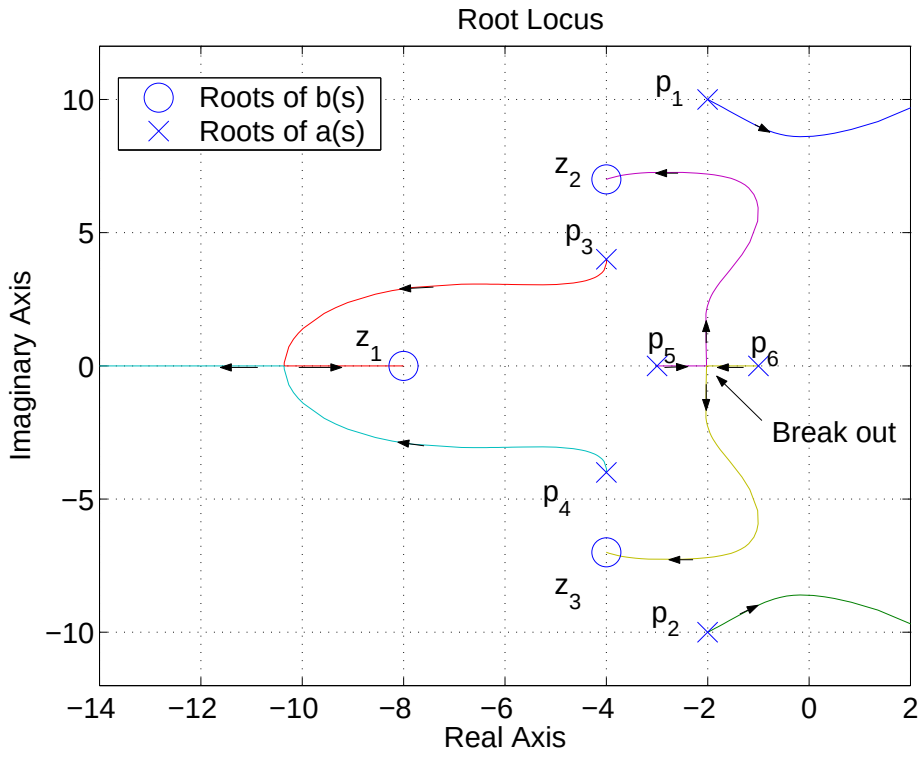


Figure 17: Example root locus showing break out.

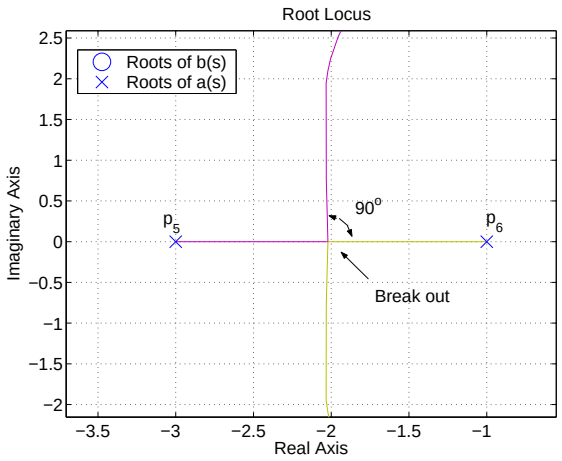


Figure 18: Blow up of figure 17.

2.2.3 Break in

- When 2 root loci come together from off the real axis; they join and break in (into the real axis); as seen in figure 19.
- The angle of arrival at a break in is $\pm 90^o$, as seen in the blow up (figure 20).

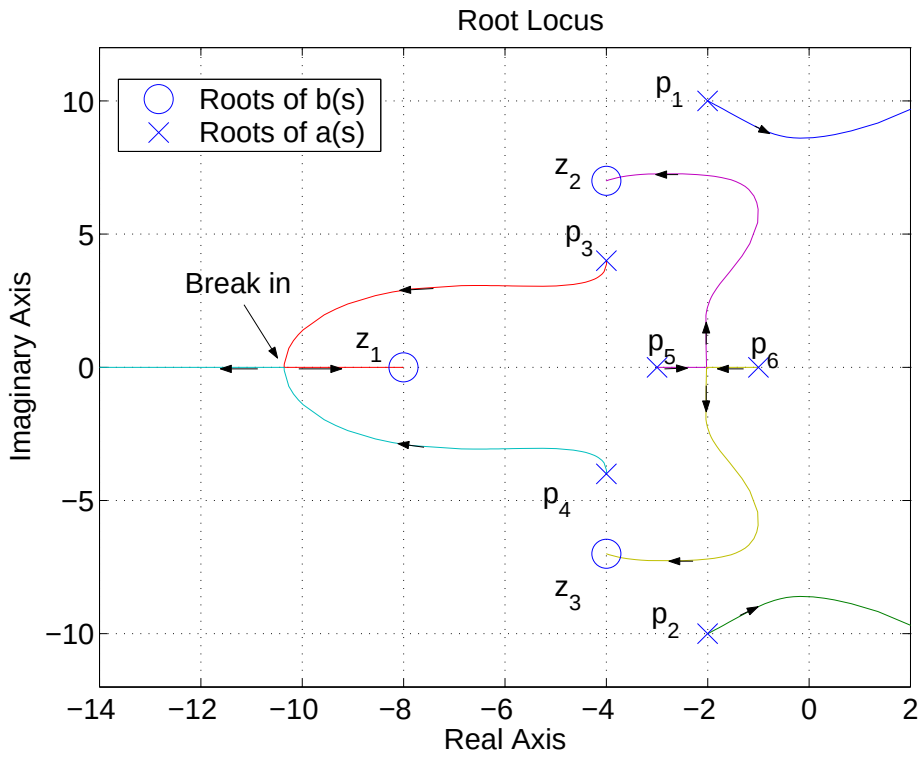


Figure 19: Example root locus showing break out.

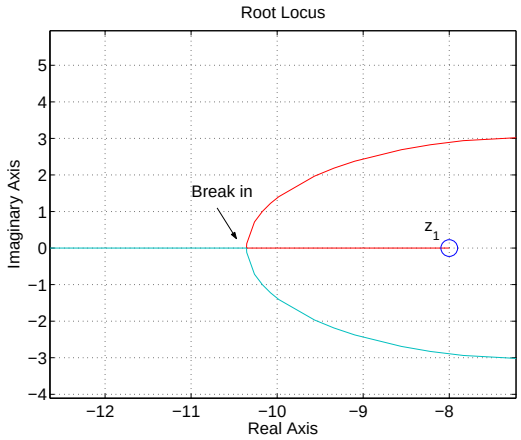


Figure 20: Blow up of figure 19.

2.2.4 Departure angle

- The departure angle is the angle at which the root locus first leaves a pole; and is illustrated as ϕ_i in figure 21.
- Each pole p_i has departure angle ϕ_i.
- This is the definition, theory presented shortly.

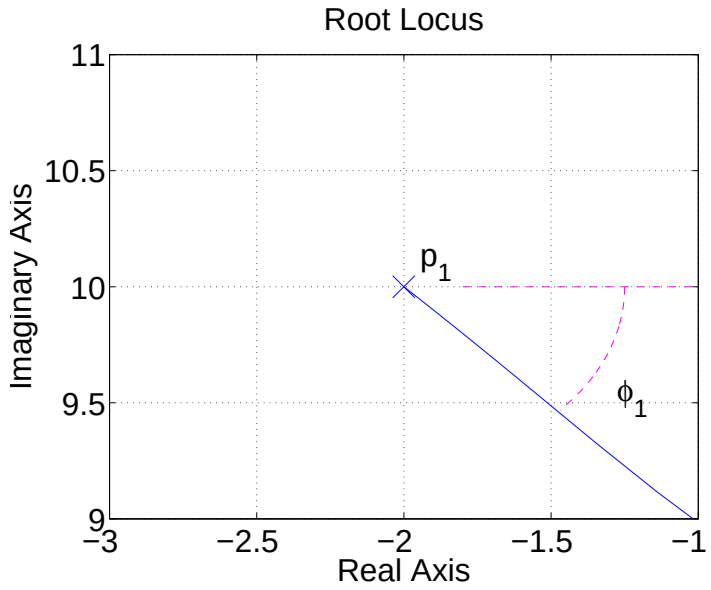
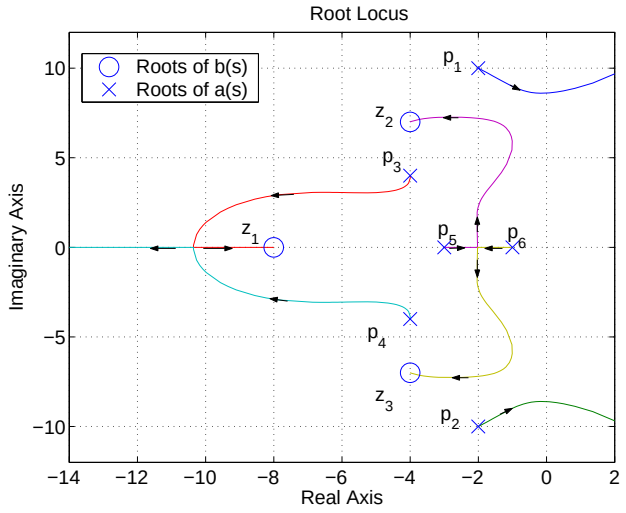


Figure 21: Example root locus illustrating departure angle.



Pole	Departure Angle
p_1	-44.8^o
p_2	44.8^o
p_3	-85.6^o
p_4	85.6^o
p_5	0°
p_6	180°

Figure 22: Departure angle values.

2.2.5 Arrival angle

- m root-locus branches "arrive" at open-loop zeros, as $K \to \infty$.
- Arrival angle is the opposite of departure angle, it is the angle ψ_j at which the root locus arrives at zero z_j.

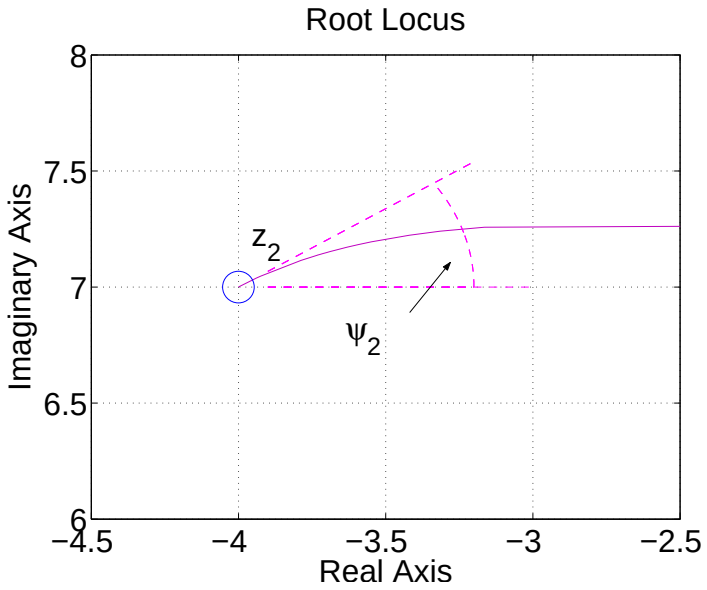
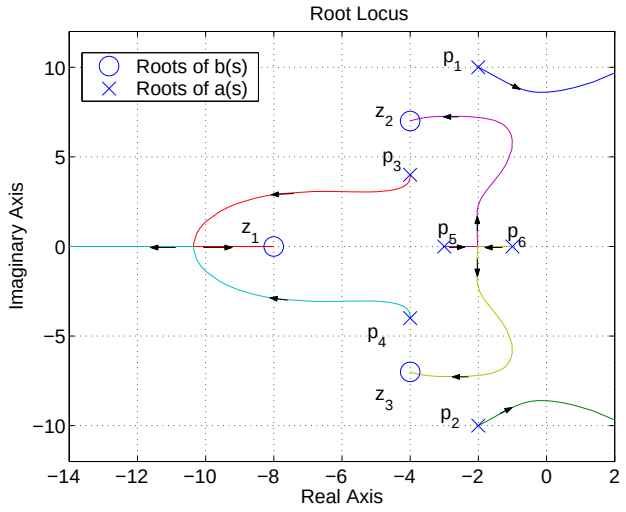


Figure 23: Example root locus illustrating departure angle.



Zero	Arrival Angle
z_1	-180.0^o
z_2	34.1^0
z_3	-34.1^0

Figure 24: Arrival angle values.

2.2.6 Asymptotes

- The root locus has n branches, they all go somewhere !
  - m root-locus branches go to zeros
  - $n-m$ root-locus branches go to ∞ along asymptotes
  - $(n-m)$ is the ***relative degree*** . Relative Degree

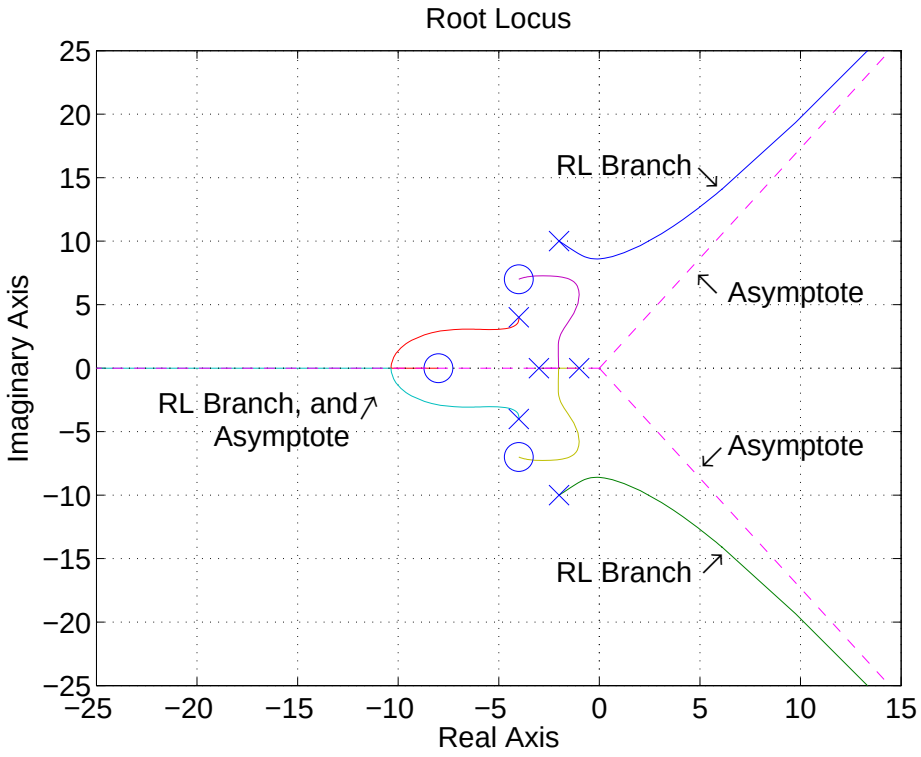


Figure 25: Example root locus showing three root-locus branches going to infinity.

- The lie of the asymptotes is easily calculated (coming up).

2.2.7 Centroid and Orientation of Asymptote branches

- The centroid is the intersection point of the asymptotes. It is given by:

$$\alpha_c = \frac{\sum_{i=1}^{n} p_i - \sum_{j=1}^{m} z_j}{n-m} = \frac{\text{Sum of OL Poles} \; - \; \text{Sum of OL Zeros}}{\text{Relative degree}}$$

- There are $n-m$ asymptote branches, their angles are given by:

$$\phi_l = \frac{180^o + 360^o\,(l-1)}{n-m}, \quad l = 0 \cdots n-m-1 \tag{18}$$

- But it is easier to remember the asymptote angles by their patterns:

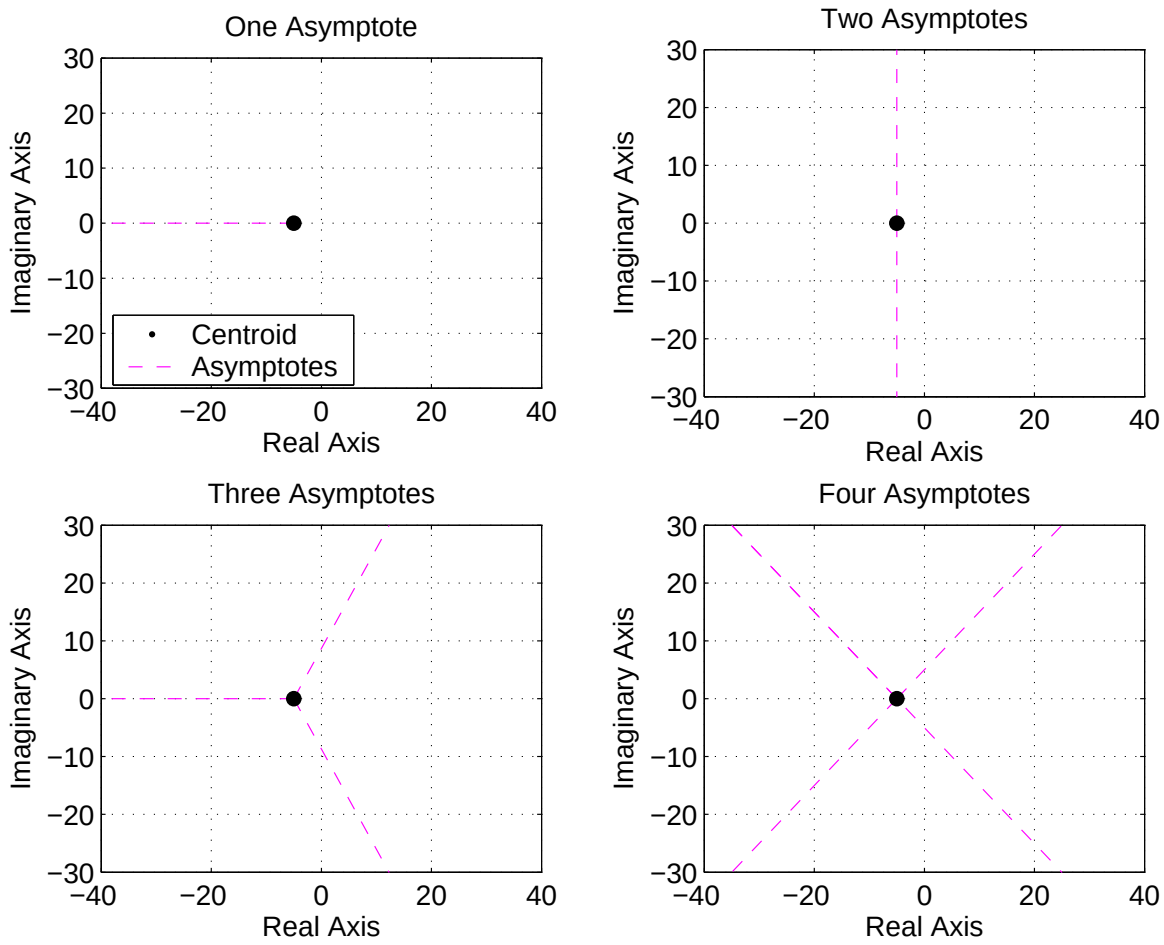


Figure 26: Example root locus showing three root-locus branches going to infinity.

2.2.8 Frequency of crossing the imaginary axis

- The frequency where a root-locus branch first crosses the $j\omega$ axis will be the oscillation frequency as the gain is increased to the point of instability.
  - It can also be determined approximately from a sketched root-locus plot.
  - It can also be determined precisely from a plot of the root locus made with Matlab.

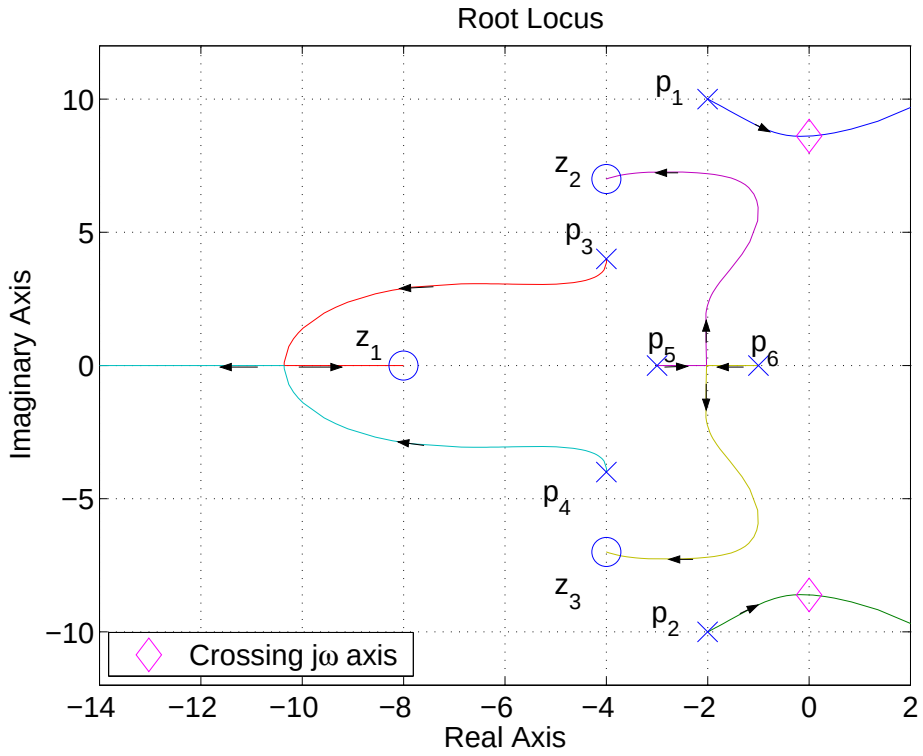


Figure 27: Example root locus showing eight distinct phenomena.

3 Five rules to sketch the root-locus (following Franklin *et al.* Chap 5)

- Five rules for sketching the root-locus are outlined here. Mathematical demonstrations of the rules are provided in section 3.4 and Franklin *et al.*

3.1 Statement of the rules (mathematical derivations follow)

Rule_1: The n root-locus branches start at the n poles of $G_L(s)$,

- m of these branches go to the m zeros of $G_L(s)$

Rule 2: Segments of the real axis to the left of an odd number of poles and zeros are in the root-locus.

Rule 2.a: The root locus is symmetric about the real axis: Poles come in complex conjugate pairs.

Rule 3: $(n-m)$ root-locus branches follow asymptotes to ∞. Asymptotes have a centroid and pattern.

1. The number of asymptotes is $(n-m)$,
2. The centroid is given by:
$$\alpha_c = \frac{\sum OL\,poles \; - \sum OL\,zeros}{n-m} \tag{19}$$
3. The asymptote angles are given by:
$$\phi_{Asymp} = \frac{180 + 360(l-1)}{n-m} \;\; l = 1, 2, \cdots \; (n-m) \tag{20}$$

Rule 4: Angles of departure from and arrival to complex poles and zeros

- For non-repeated comlex poles, the angle the root-locus branch makes departing from the pole is:
$$\phi_{k,dep} = \sum_{j=1}^{m} \psi_j - \sum_{\substack{i=1 \\ i \neq k}}^{n} \phi_i - 180^o \tag{21}$$

  Where ψ_j : angles to zeros, ϕ_i: angles to poles.
  Arrival angle of branch to complex zero has a similar equation (below).

Rule 5: Angles of root-locus branches at points where they intersect

- Points of break-out and break-in are at $\pm 90^o$.
- Points of break-out from the real axis, between two real poles, are approximately half-way between the poles.
- Points of break-in to the real-axis, between two real zeros, are approximately half-way between the zeros.

3.2 Procedure to sketch the root locus

Step 1. Draw the open-loop pole-zero constellation. Apply rule 1 and determine the number of root-locus branches.

Step 2. Apply rule 2, and determine which segments of the real axis are in the root locus.

Step 3. Apply rule 3, and determine the centroid and asymptotes

Step 4. Considering the segments on the real axis, and applying rule 5, identify the points of break-out and break-in.

Step 5. (Sketching) Sketch the root locus. Connect the open-loop poles with

1. Open-loop zeros, or

2. Asymptotes

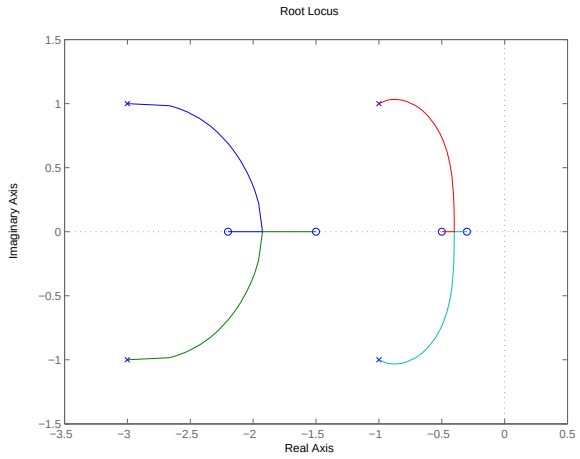
- For sketching, it is rarely necessary to calculate departure and arrival angles.

- General rule: Think of the root-locus branches as springs. Draw in root-locus branches to connect the segments with minimum bending.

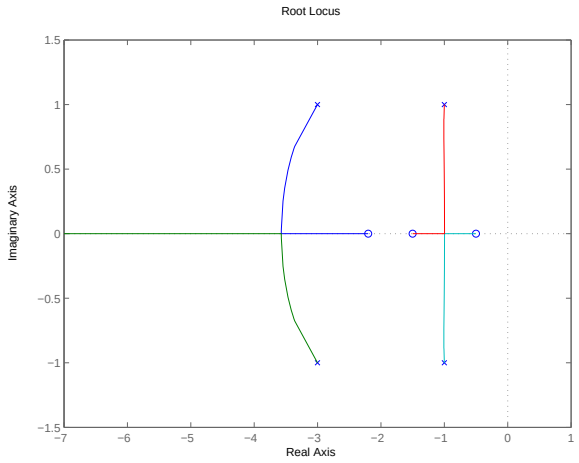
3.3 Root locus examples

- These examples show the root loci of several pole-zero constellations
- In each the number of asymptotes is

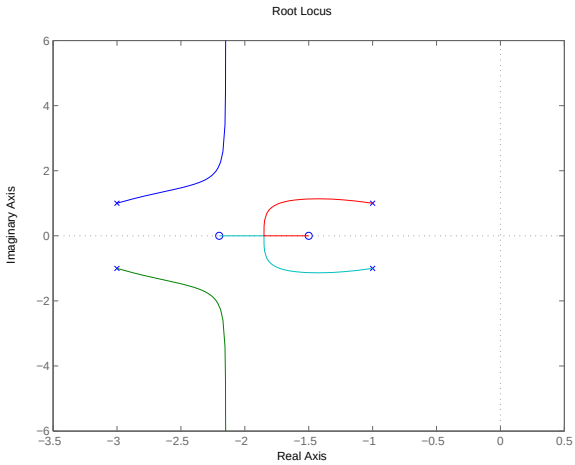
$$N_a = n - m$$



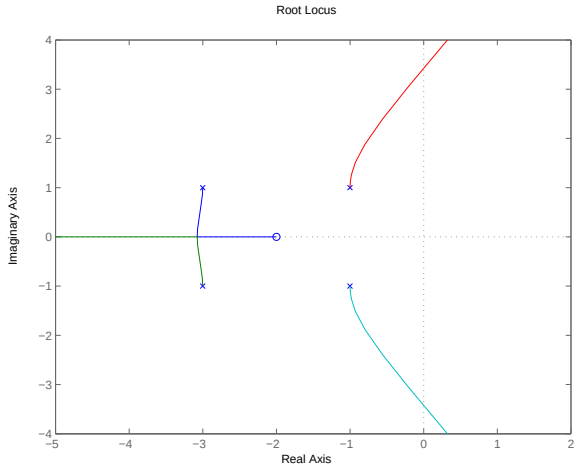
(a) Four poles, four zeros (0 Asymptotes)



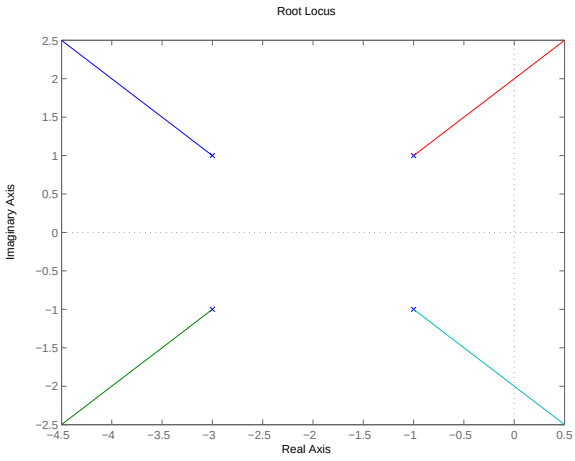
(b) Four poles, three zeros (1 Asymp.)



(c) Four poles, two zeros (2 Asymp.)



(d) Four poles, one zero (3 Asymp.)



(e) Four poles, no zeros (4 Asymp.)

- Matlab's root locus command draws the root loci, and with a mouse click can indicate properties

```
GL =  tf(1 3], [1 4 5 0]);
figure(2), clf
 rlocus(GL)
  SetLabels(13)
  print('-deps2c', 'RLocus_Angle')
```

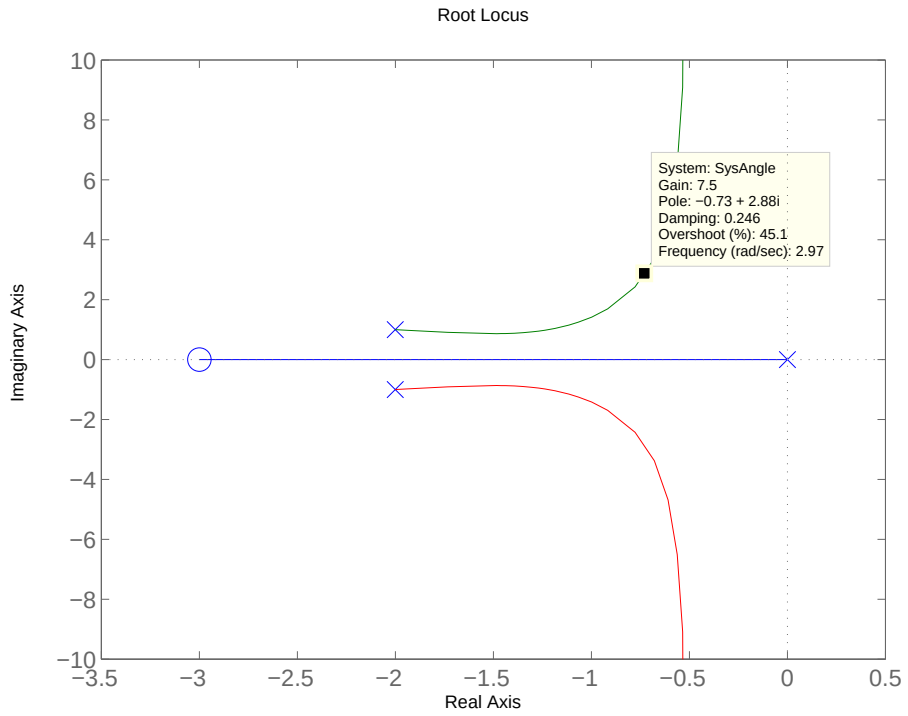


Figure 28: Root locus for the aircraft pitch angle, P-type control.

3.4 Mathematical derivations of the rules for sketching the root locus

Rule_1: The n root-locus branches start at the n poles of $G_L(s)$, m of these branches go to the m zeros of $G_L(s)$

- Recall the characteristic equation can be written:

$$a(s) + K_{rlg}\, b(s) = 0 \tag{22}$$

with

$$K_{rlg}\frac{b(s)}{a(s)} = K_{rlg}\frac{s^m + b_{m-1}s^{m-1}\cdots + b_1 s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0}$$

When $K_{rlg} >> 1$ Eqn (22) has two approximations

1. When $s << K_{rlg}^{\frac{1}{n-m}}$, then $a(s) << K_{rlg}\, b(s)$, and Eqn (22) is approximated by

$$a(s) + K_{rlg}\, b(s) \simeq K_{rlg}\, b(s) = 0$$

which places m of the closed-loop poles near the roots of $b(s)$, which are the open-loop zeros.

2. When $s >> K_{rlg}^{\frac{1}{n-m}}$, Eqn (22) is approximated by the highest powers in s:

$$a(s) + K_{rlg}\, b(s) \simeq s^n + K_{rlg}\, s^m = s^m\left(s^{n-m} + K_{rlg}\right) \tag{23}$$

$s^{n-m} + K_{rlg} = 0$ has roots at:

$$s_i = K_{rlg}^{\frac{1}{n-m}} \angle \frac{180 + 360(l-1)}{n-m} \tag{24}$$

As $K_{rlg} \to \infty$, $n-m$ closed-loop poles go to infinity, along the asymptotes given by Eqn (24).

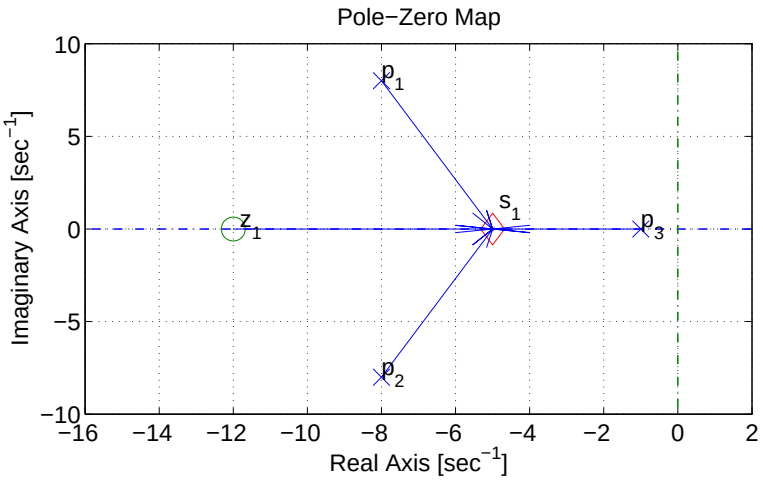
Rule 2: Segments of the real axis to the left of an odd number of poles and zeros are in the root-locus.

- Recall the Angle Condition: for a value of s_o in the root locus:

$$\begin{aligned} \angle G_L(s_0) &= \sum_{i=1}^{m} \text{Angles from Zeros to } s_0 - \sum_{i=1}^{n} \text{Angles from Poles to } s_0 \\ &= 180 + 360(l-1) \end{aligned} \tag{25}$$

  - Consider the two examples of figure 29

(A)



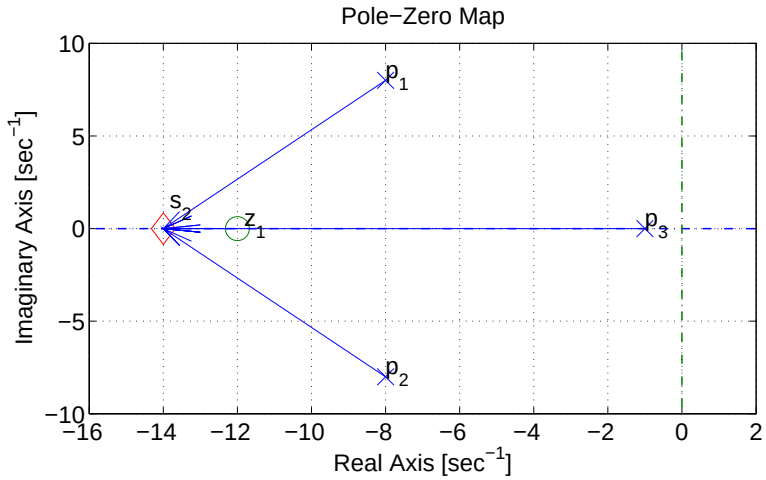
$\angle$ from p_1: -60°

$\angle$ from p_2: $+60^o$

$\angle$ from p_3: 180°

$\angle$ from z_1: 0°

(B)



$\angle$ from p_1: -135°

$\angle$ from p_2: $+135^o$

$\angle$ from p_3: 180°

$\angle$ from z_1: 180°

Figure 29: Vectors from poles and zeros to an example point on the real axis.

In figure 29(A),

$$\angle G_L(s_1) = 0^o - (-60^o + 60^o + 180^o) = -180^o \tag{26}$$

In figure 29(B),

$$\angle G_L(s_2) = 180^o - (-60^o + 60^o + 180^o) = 0^o \tag{27}$$

The examples illustrate the general principles:

- When s_i is on the real axis, the angles from complex pole or zero pairs always cancel
- When s_i is to the left of a pole, an angle of -180° is introduced into $\angle G_L$
- When s_i is to the left of a zero, an angle of $+180^o$ is introduced into $\angle G_L$

Considering these principles, in order for $\angle G_L(s_i) = 180^o + 360^o(l-1)$, s_i must be to the left of an odd number of poles and zeros.

- Complex conjugate pairs of poles or zeros cancel out (but also don't change the number between even and odd).
- When s_i is to the right of poles or zeros, the contributed angle is 0°.
- An odd number of $+180^o$ and -180° adds up to $\pm 180^o$.

Rule 2.a: The root locus is symmetric about the real axis.

- Poles come in complex conjugate pairs.

Rule 3: $(n-m)$ root-locus branches follow asymptotes to ∞. Asymptotes have a centroid and pattern.

1. The number of asymptotes is $(n-m)$,
2. The centroid is given by:

$$\alpha_c = \frac{\sum poles \; - \sum zeros}{n-m} \tag{28}$$

3. The asymptote angles are given by:

$$\phi_{Asymp} = \frac{180 + 360(l-1)}{n-m} \;\; l = 1, 2, \cdots \; (n-m) \tag{29}$$

- Recall from the demonstration of rule 1, when $K_{rlg} >> 1$, the characteristic equation is approximated by

$$a(s) + K_{rlg}\, b(s) \simeq s^m \left(s^{n-m} + K_{rlg} \right) = 0$$

has roots at:

$$s_i = K_{rlg}^{\frac{1}{n-m}} \angle \frac{180 + 360\,(l-1)}{n-m} \tag{30}$$

Eqn (30) gives the angle condition.

- (Following Franklin *et al.* section 5.2.1 (7th ed.)) The centroid, α_c, is determined by considering that a monic polynomial is equal to the product of its factors

$$a(s) = s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_o = (s-p_1)(s-p_2)\cdots(s-p_n) \tag{31}$$

Multiplying out the right-hand side gives

$$\begin{aligned} -a_{n-1} &= \sum_n p_i \\ -b_{m-1} &= \sum_m z_i \end{aligned}$$

where the second equation comes from applying Eqn (31) to the zeros polynomial.

- From $a(s) + K_{rlg}\, b(s) = 0$, the characteristic equation is written

$$s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_o + K_{rlg}\left(s^m + b_{m-1}\, s^{m-1} \cdots + b_1 s + b_0\right) = (s - r_1)(s - r_2)\cdots(s - r_n) = 0 \tag{32}$$

where the r_i are the closed-loop poles.

- Notice that the sum of the closed loop poles is the negative of the coefficient of the s^{n-1} term, and is independent of K_{rlg} if $m < n-1$. Therefore, if $G_L(s)$ has at least two more poles than zeros, $a_1 = -\sum r_i$, so it follows that

$$\sum_n r_i = \sum_n p_i$$

We have previously seen that m of the root-locus branches go the m zeros. Breaking the sum of r_i into two parts, according to whether the root is on an asymptote or or goes to a zero

$$\sum_{\text{Asymptotes}} r_i + \sum_{\text{zeros}} r_i = \sum p_i \tag{33}$$

- For the branches that go to zeros, $\sum_{\text{zeros}}\ r_i = \sum z_i$, so Eqn (33) can be written:

$$\sum_{n-m} r_i = \sum_n p_i - \sum_m z_i \tag{34}$$

- The centroid is the average of the locations of the poles on the asymptotic branches, giving:

$$\alpha_c = \frac{\sum_{n-m} r_i}{n-m} = \frac{\sum_n p_i - \sum_m z_i}{n-m} \tag{35}$$

- Eqns (35) and (30) establish the centroid and angles conditions of rule 3.

Rule 4: Angles of departure from and arrival to complex poles and zeros

- The rules for arrival and departure angle are complicated by the possibility of repeated poles and zeros. In the following q is the multiplicity of a repeated pole (or zero). $q = 1$ for single poles, $q = 2$ for double poles, etc. l is the index on the repeated pole (or zero).

- The usual case is single poles (or zeros), in which case $q = 1$, $l = 1$.

4.a The departure angle of a root-locus from the kth pole is given by:

$$q\,\phi_{k,l,dep} = \sum_{j=1}^{m} \psi_j - \sum_{\substack{i=1 \\ i\neq k}}^{n-q} \phi_i - (180^o + 360^o\,(l-1)) \tag{36}$$

Where: q is the multiplicity of the kth pole

$\phi_{k,l,dep}$ is the departure angle of the kth branch from the open-loop pole, l is the index on repeated poles

$\sum \psi_i$ is the sum of the angles from zeros to the pole

$\sum \phi_i$ is the sum of the angles from other poles to pole k.

- In the case where the pole is not repeated, $q = 1$, Eqn (36) simplifies to

$$\phi_{k,dep} = \sum_{j=1}^{m} \psi_j - \sum_{\substack{i=1 \\ i\neq k}}^{n} \phi_i - 180^o \tag{37}$$

4.b The arrival angle of a root-locus at zero l is given by:

$$q\,\psi_{k,arr} = \sum_{j=1}^{n} \phi_j - \sum_{\substack{i=1 \\ i\neq j}}^{m-q} \psi_i + 180^o + 360^o\,(l-1) \tag{38}$$

Where: q is the multiplicity of the kth zero

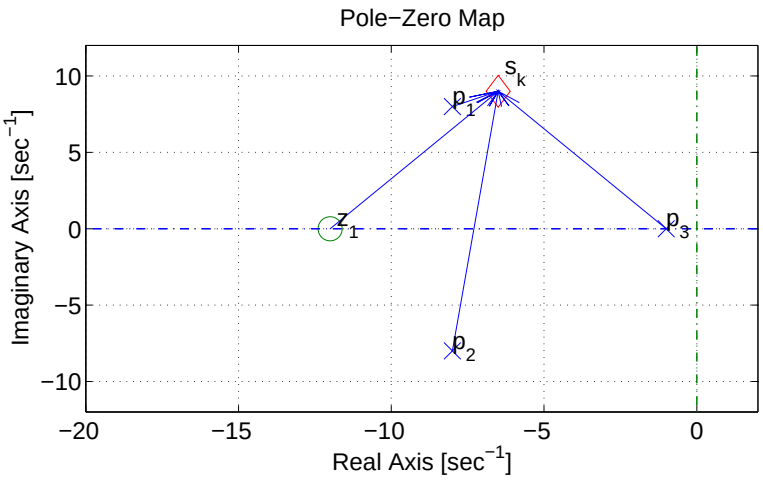
$\psi_{k,l,dep}$ is the arrival angle of the lth branch to open-loop zero k, l is the index on repeated zeros

$\sum \phi_j$ is the sum of the angles from poles to the zero

$\sum \psi_i$ is the sum of the angles from other zeros to zero k.

- In the case where the zero is not repeated, $q = 1$, Eqn (38) simplifies to

$$\psi_{k,arr} = \sum_{j=1}^{n} \phi_j - \sum_{\substack{i=1 \\ i\neq j}}^{m} \psi_i + 180^o \tag{39}$$



$\angle$ from p_1: -60°

$\angle$ from p_2: $+60^o$

$\angle$ from p_3: 180°

$\angle$ from z_1: 0°

Figure 30: Vectors from poles and zeros to an example point on the real axis, repeated.

- To demonstrate rule 4, consider the case of figure 30. Since test point s_k is near to pole p_k, the angles to the test point are approximately the angles to pole p_k. The angle condition becomes

$$\begin{aligned}\angle G_L(s_k) &= \sum_{i=1}^{m} \text{Angles from Zeros to } s_k - \sum_{i=1}^{n} \text{Angles from Poles to } s_k \\ &= 180 + 360(l-1)\end{aligned}$$

$$= \sum_{i=1}^{m} \text{Angles from Zeros to pole } p_k - \sum_{i=1}^{n-1} \text{Angles from other Poles to pole } p_k$$
$$-\text{Angle from } p_k \text{ to } s_k \qquad (40)$$

- Eqn (40) can be solved for $\phi_{k,l,dep}$, giving Eqn (36).

- The same argument leads to Eqn (38) for arrival angle of a branch to a zero.

Rule 5: Angles of root-locus branches at points where they intersect

At points s' where q branches intersect (such as points of break-out and break-in, where $q = 2$), the branches will approach the point s' at angles separated by:

$$\theta = \frac{180^0 + 360^o\,(l-1)}{q} \tag{41}$$

- To demonstrate rule 5, introduce the ***continuation locus***. We have considered $0 \leq K_{rlg} \leq \infty$. But now lets break then range into two segments:

  1. Let the original K_{rlg} vary on the range

  $$0 \leq K_{rlg} \leq K_1$$

  2. Introduce a new model, with $K'_{rlg} = K_1 + K_2$. The new model has a characteristic equation of:

  $$a(s) + K_1 b(s) + K_2 b(s) = 0$$

- Define $a'(s) = a(s) + K_1 b(s)$, we can continue the root locus in the standard way, working with characteristic equation

$$a'(s) + K_2 b(s) = 0$$

For the continuation root locus, the closed-loop poles at $K_{rlg} = K_1$ become the open-loop poles of the continuation locus.

- If we choose K_1 to be the value of K_{rlg} where the root-locus branches intersect, then in the continuation locus we have a repeated open-loop pole at that location, and the angle of departure equation, Eqn (36), applies. Eqn (36) gives a separation angle of Eqn (41).

- Practical significance: for $q = 2$, rule 5 gives a separation angle of 180°, meaning that branches break out of and break in to the real axis at angles of $\pm 90^o$.

3.5 Discovery of the root locus



Figure 31: Walter Evans as a graduate student.

- The root locus was devised by Walter R. Evans in 1948.

  W.R. Evans, "Control system synthesis by root locus method," *Transactions of the AIEE*, vol. 69, pp. 66–69, 1950.

- Evans was a graduate student at UCLA and teaching a course in controls at the North American Aviation Co. (Now part of Boeing). A student (now anonymous) asked a question about an approximation, leading to discovery of the mathematical foundations of the root-locus method.

- Controller design comprises two stages:

  - Selection of the controller structure

  - Tuning of the controller parameters

  The root locus method aids with each stage.

4 Root locus design example

- Controller design comprises
  1. Selecting the controller structure
  2. Tuning the numerical controller parameters
- Sketching the root locus gives us a way to assess the possibilities of various controller structures
- Standard controller structures are listed in table 1.

P	PD	PI	PID
Proportional	Proportional-Derivative	Prop.-Integral	Prop.-Int.-Deriv.
$G_c(s) = k_p$	$G_c(s) = k_d s + k_p$	$G_c(s) = \frac{k_p s + k_i}{s}$	$G_c(s) = \frac{k_d s^2 + k_p s + k_i}{s}$

Phase Lead	Phase Lag	Lead-Lag
$G_c(s) = K_c \frac{(s+z_1)}{(s+p_1)}$, $\lvert z_1 \rvert < \lvert p_1 \rvert$	$G_c(s) = K_c \frac{(s+z_1)}{(s+p_1)}, \lvert z_1 \rvert > \lvert p_1 \rvert$	$G_c(s) = K_c \frac{(s+z_1)(s+z_2)}{(s+p_1)(s+p_2)}$

Table 1: Standard controller structures.

- Common applications for standard controller structures

  PI: Velocity control of motor drives, temperature control (good accuracy, acceptable stability and speed)

  PD: Position control where high accuracy is not required (good speed and stability, so-so accuracy)

  PID: Used many many places. Controls scientist Karl has estimated that 80% of single-loop controllers in service are PID
  (good, speed accuracy, stability)

  Phase-lead: Similar to PD, but less amplification of high-frequency noise (3 params. not 2).

  Phase-lag: Similar to PI. . Used where a pole at the origin is unacceptable (3 params. not 2).

  Lead-lag: Similar to PID. Used where a pole at the origin is unacceptable, can be as good as PID (notice, 5 parameters rather than 3)

4.1 First design example

- Consider the system of figure 32, with $$G_p(s) = \frac{40}{(s+2)(s+6)(s+6)}$$

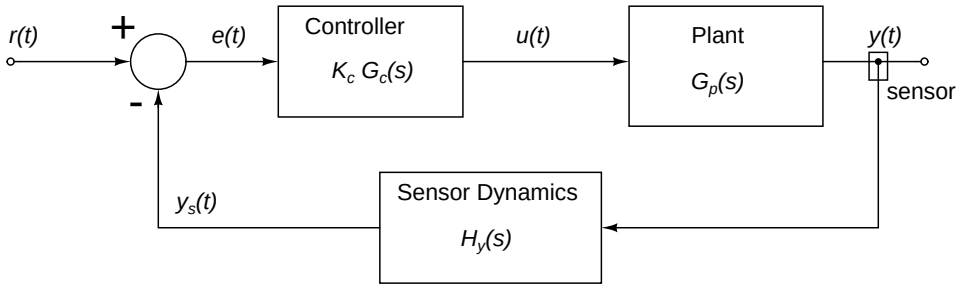
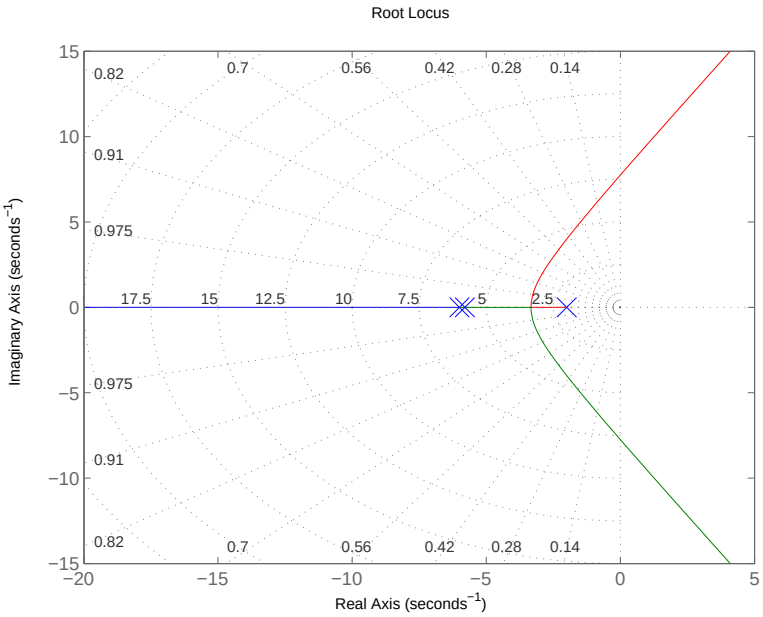


Figure 32: Block diagram,

4.1.1 Selecting the controller structure

- P-type control is simplest. Let's consider the possibilities of p-type control




```
Gp = tf(40, poly([-2, -6, -6]))
GcP = tf(1, 1);
figure(1),
   rlocus(GcP*Gp)
   title('P-type control')
```

Figure 33: $G_p(s)$ with p-type control.

- The system has 3 asymptotes
- The system is lightly damped for $k_p = 3$, and unstable at $k_p = 19$.

P-type control

- k_p= 1 gives a balance between speed, accuracy and damping.

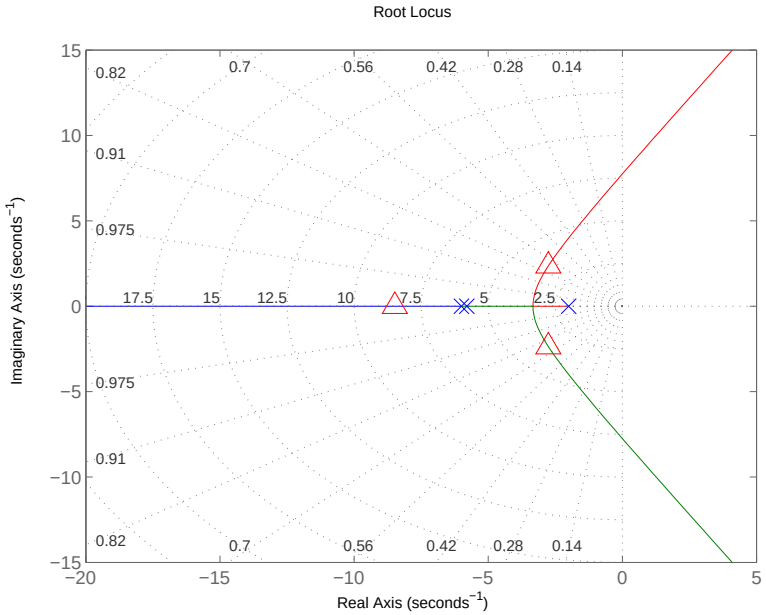


Figure 34: Root locus of $G_p(s)$ with p-type control, with $k_p = 1$ and closed-loop poles shown.

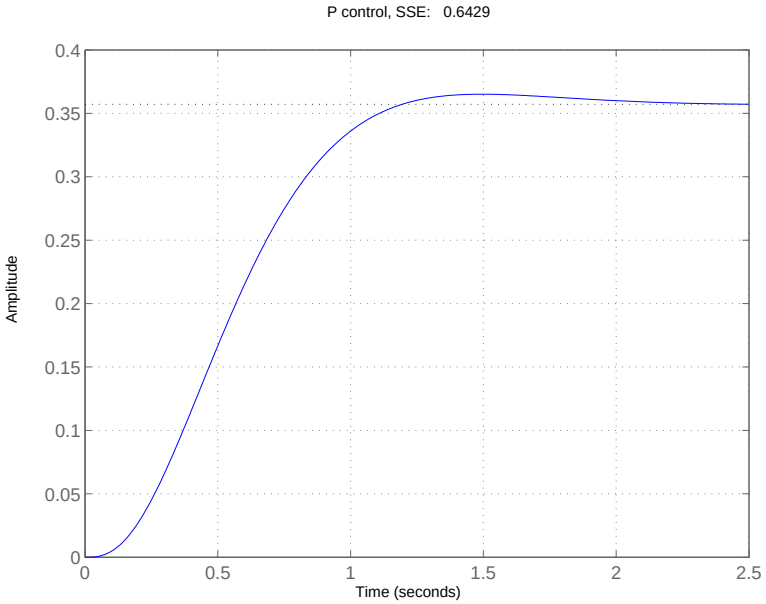


Figure 35: Step response of $G_p(s)$ with p-type control.

PD Control

- Now consider PD control, with

$$K_c G_c(s) = k_d s + k_p = K_c (s + \alpha)$$

  where $-\alpha$ is the zero location of the PD controller.

- First try putting the zero above the double pole at $s = -6$, with $z = -10$.

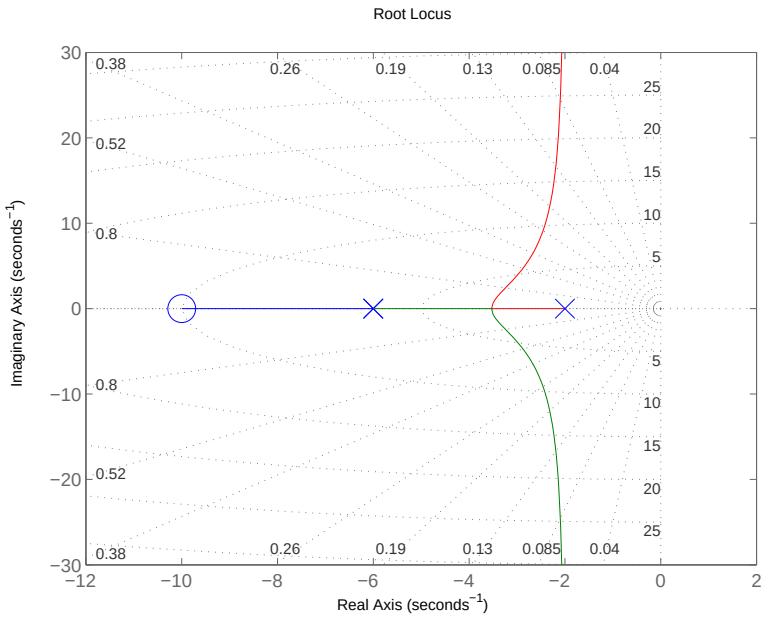


Figure 36: Root locus of $G_p(s)$ with PD-type control, and the zero at $z = -10$.

- The root locus shows that the system can not be made very fast.

- Let's try moving the zero into the space between the poles.

PD Control, try another position for the zero

- Try putting the zero below the double pole at $s = -6$, with the zero at $z = -5$.

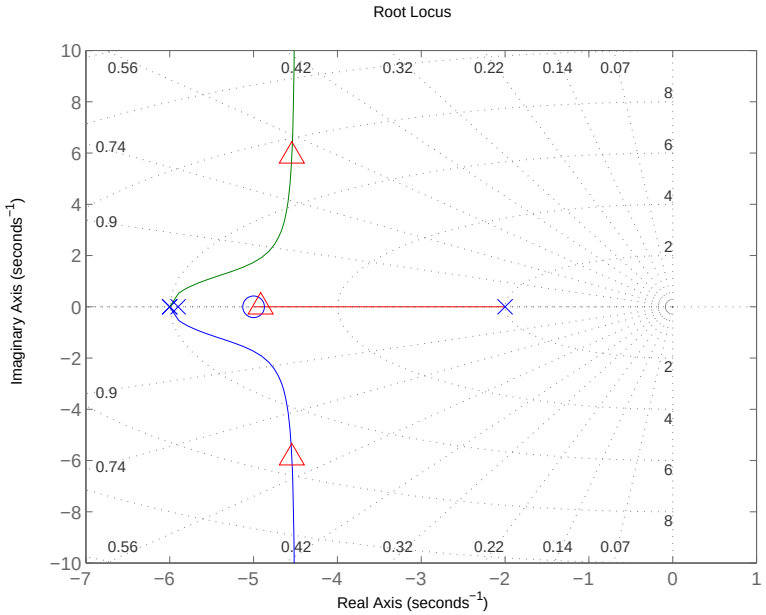


Figure 37: Root locus with PD control, and the zero at $z = -5$ and $K_c = 1$.

- The root locus shows that the system is more than twice as fast as previously.

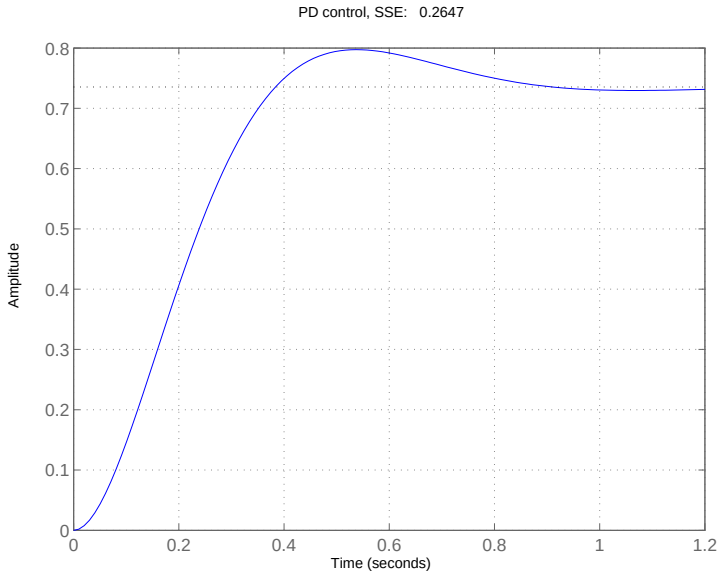


Figure 38: Step response of $G_p(s)$ with PD control, compare with figure 35.

Consider PI control PI control has 2 parameters, it gives the freedom to locate the zero and choose K_c , the over-all controller gain

$$G_c(s) = \frac{k_p s + k_i}{s} = K_c \frac{s+\alpha}{s} \tag{42}$$

Put the zero just above the pole at $s = -2$, to try to bring the pair of branches to the left.

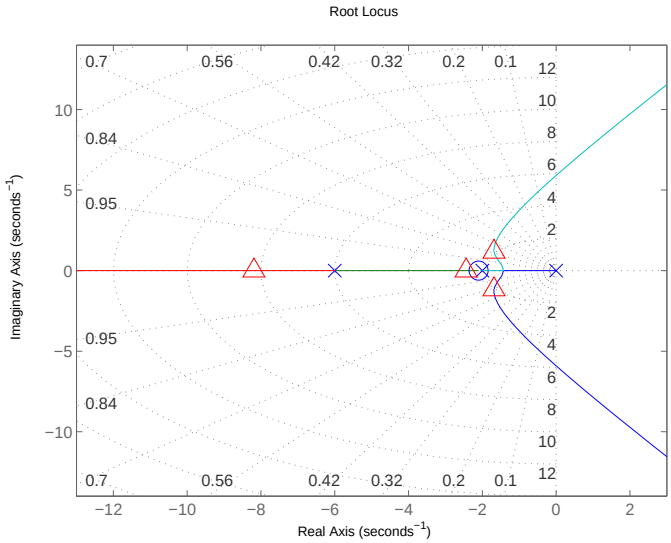


Figure 39: Root lo and $K_c = 1$.

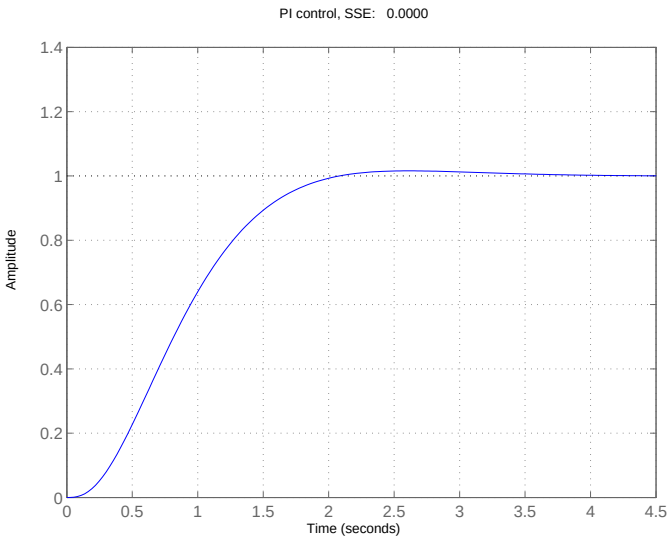
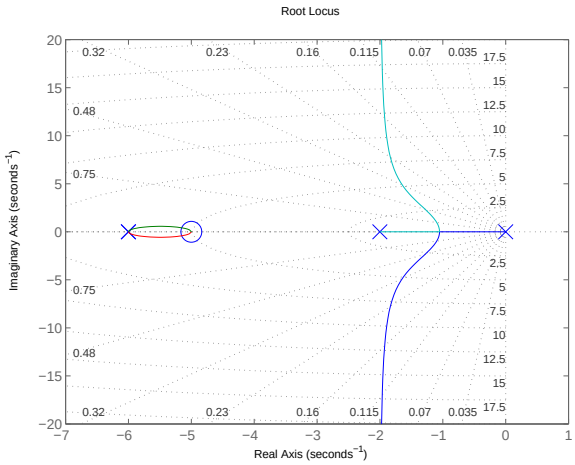


Figure 40: Step response of $G_p(s)$ with PI control.

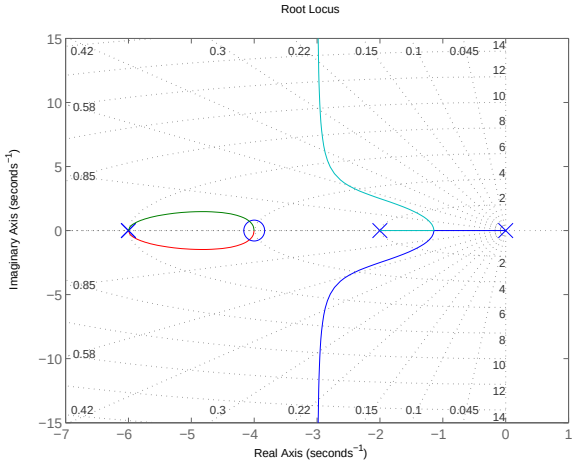
- PI Control gives zero steady-state error, but is not very fast.

Consider PID control

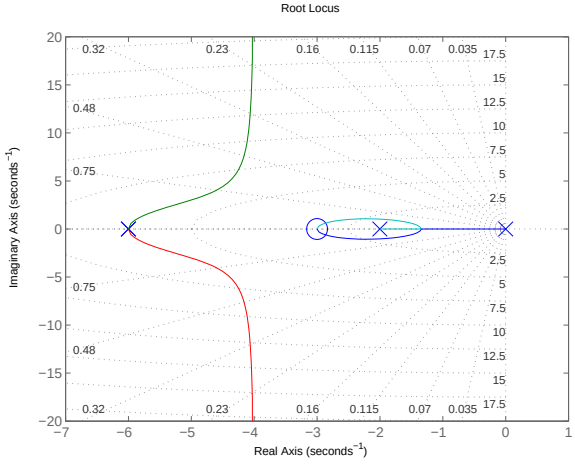
$$G_c(s) = \frac{k_d s^2 + k_p s + k_i}{s} = K_c \frac{(s - z_1)(s - z_2)}{s} \tag{43}$$



$z = -5$



$z = -4$



$z = -3$

Figure 41: $G_p(s)$ with PID-type control, and zero locations.

Consider PID control

- Choose the two compensator zeros at $s = -3$, to bring the low-speed branches around.

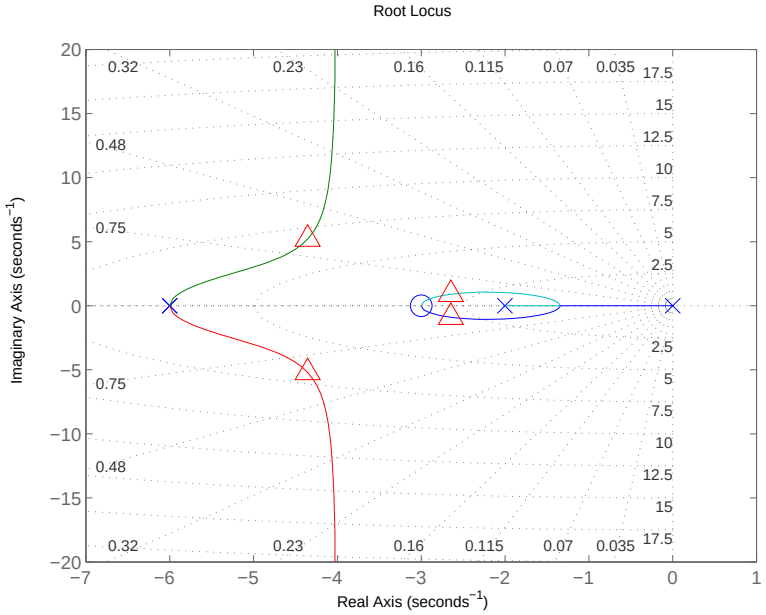


Figure 42: Root [illegible] d $K_c = 1$.

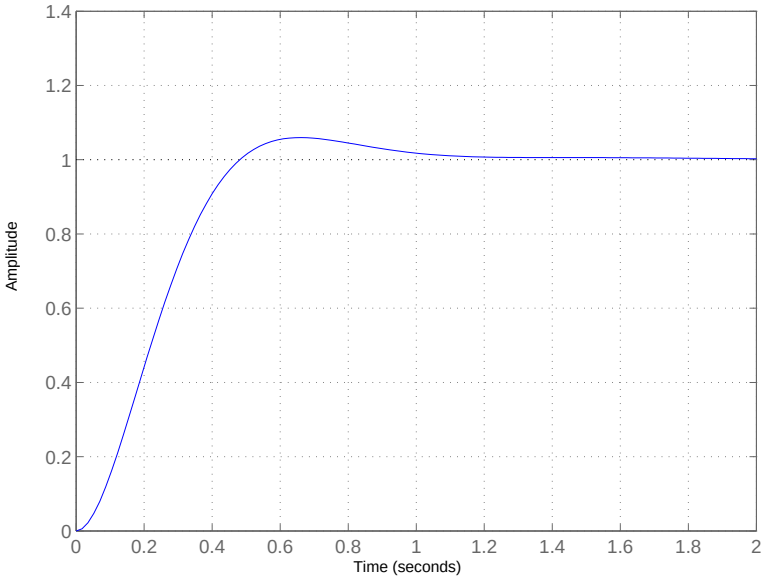


Figure 43: Step response of $G_p(s)$ with PI control.

- PID Control gives zero steady-state error, and is faster.

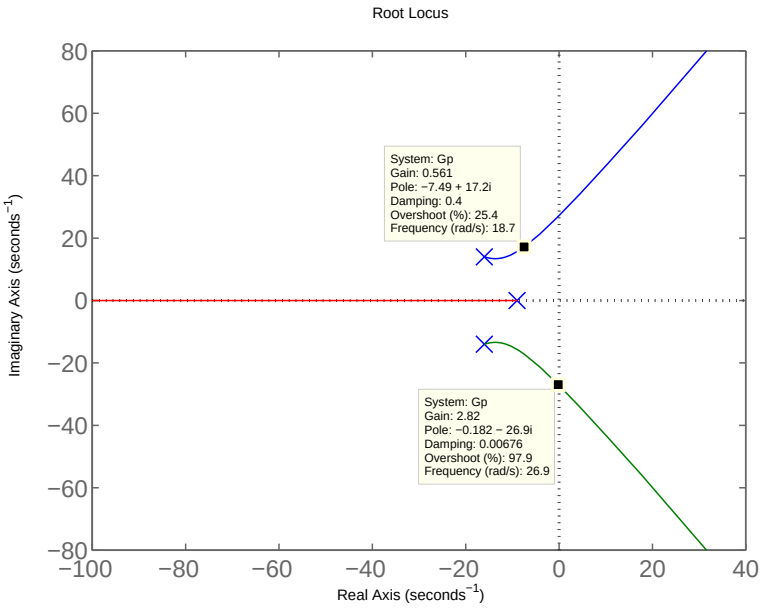
4.2 Second design example

- Consider the system of figure 32 (above), with

$$G_p(s) = \frac{9000}{(s+9)\,(s+16 \pm j\,14)}$$

4.2.1 Selecting the controller structure

- P-type control is simplest. Let's consider the possibilities of p-type control




```
Gp = tf(9000,
  poly([-9 -16+j*14, -16-j*14]))
figure(2), clf
  rlocus(Gp)
  SetLabels(16)
  print('-deps2c', 'GpP_RootLocus');
```

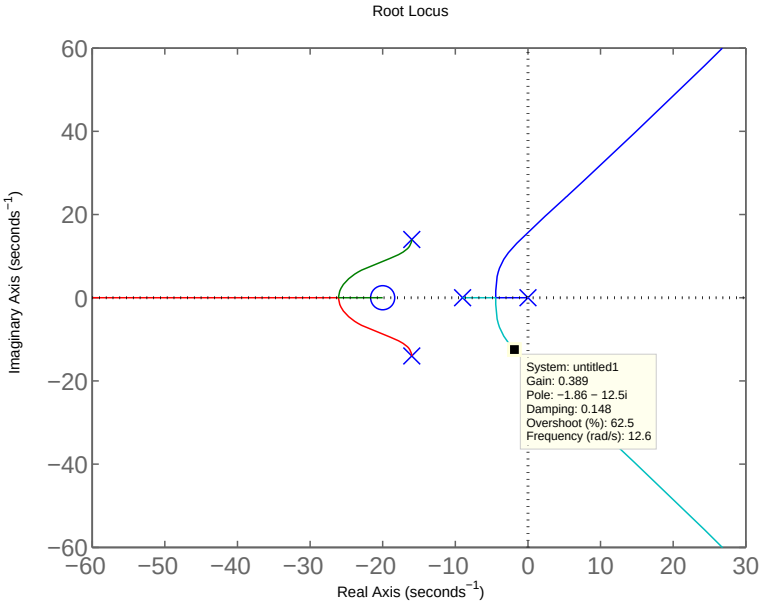
Figure 44: $G_p(s)$ with p-type control.

- The system has 3 asymptotes

- The system is lightly damped for $k_p = 0.9$, and unstable at $k_p = 2.7$.

- Consider PI control
  - PI control has 2 parameters, it gives the freedom to locate the zero and choose K_c , the over-all controller gain

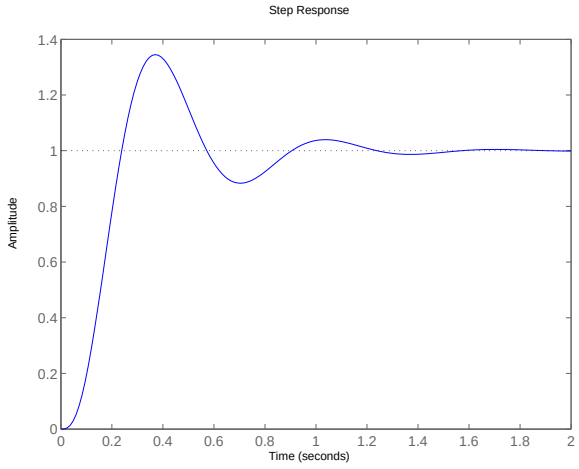
$$G_c(s) = \frac{k_p s + k_i}{s} = K_c \frac{s - z}{s} \qquad (44)$$




```
Gc = tf([1 20], [1 0])
figure(3), clf
  rlocus(Gc*Gp)
  SetLabels(16)
  print('-deps2c', 'GpPI_RootLocus');
```

Figure 45: $G_p(s)$ with PI-type control.

- The system still has 3 asymptotes
- There is now a point of break-out, and a point of break-in
- It is not possible to make the system faster than $\tau_s \sim 2$ seconds.

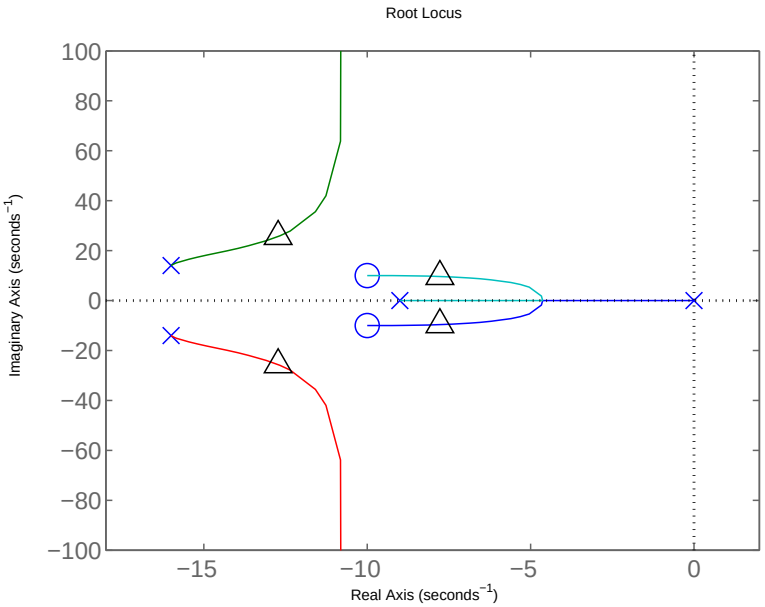



```
step(feedback(.23*Gc*Gp, 1))
  SetLabels(12)
  print('-deps2c', 'GpPI_Step');
```

Figure 46: $G_p(s)$ with PI-type control.

- Consider PID control

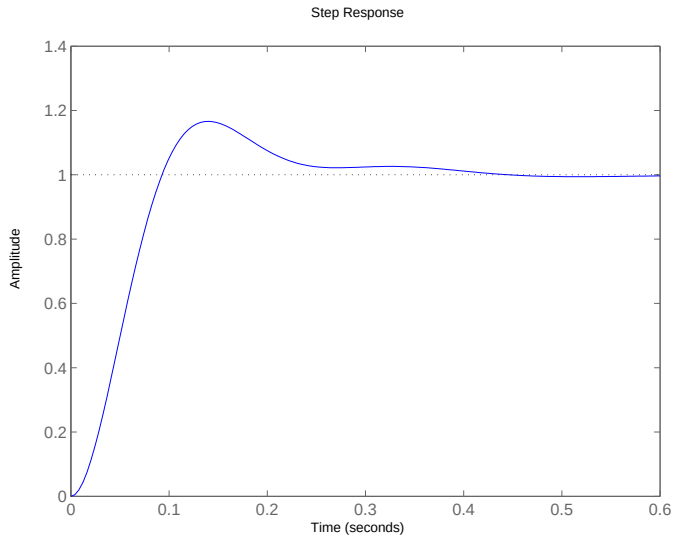
$$G_c(s) = \frac{k_d s^2 + k_p s + k_i}{s} = K_c \frac{(s - z_1)(s - z_2)}{s} \tag{45}$$




```
z1 = -10+j*10
Gc = tf(poly([z1 z1']), [1 0])
figure(3), clf
  rlocus(Gc*Gp)
  Try  = feedback(.07*Gc*Gp, 1)„
  Poles = pole(Try)
  hold on;
     plot(real(Poles), imag(Poles), 'k
  hold off;
  SetLabels(12)
  print('-deps2c', 'GpPID_RootLocus');
```

Figure 47: $G_p(s)$ with PID-type control.

- The system has 2 asymptotes
- There is now a point of break-out, but no point of break-in
- The system is substantially faster than with PI control.




```
step(Try)
  SetLabels(12)
  print('-deps2c', 'GpPID_Step');
```

Figure 48: $G_p(s)$ with PID-type control.

4.2.2 Tuning controller gains

- This task is essentially complete. For the closed-loop poles in figure 47,

$$K_c G_c(s) = 0.07 \frac{s^2 + 20s + 200}{s} = \frac{k_d s^2 + k_p s + k_i}{s} \tag{46}$$

so

$$k_d = 0.07, \quad k_p = 1.4, \quad k_i = 14$$

5 Conclusions, Part 9

- The root-locus method reveals system characteristics as a parameter varies from 0 to ∞ .

- The root-locus can be used in two ways:

  - Sketching

    Based on 5 rules, we can hand sketch the root locus, to rapidly investigate various combinations of poles and zeros. This helps select the controller structure.

  - Computer drawn

    Given a controller structure and pole and zero locations, tools like Matlab can precisely draw the root locus and report parameters at any point along the locus. This helps with parameter tuning.

- The root-locus gives us a technique to understand and explore the possibilities of multiple controller structures.

  - Choosing a controller structure (P, PI, etc.) corresponds to choosing the number of poles and zeros.

  - Tuning a controller corresponds to choosing the location controller poles and zeros, and overall controller gain.

6 Conclusions, EE 474

- We started with basic modeling for dynamic systems
  - This took us into differential equations
  - And led us to transfer functions
- We learned to analyze transfer functions and many of their basic mathematical properties
- We explored the properties of systems and transfer functions, and developed basic measures of performance
  - Speed
    - Rise time
    - Settling time
    - Bandwidth
  - Accuracy
    - Steady-state error
  - Stability
    - Determination of stability
    - Damping
- We explored the most common controller structures, P, PD, PI, PID
- We explore root-locus design, a widely used approach to controller design

$$\text{Modeling} \longrightarrow \text{Analysis} \longrightarrow \text{Design}$$

- And did each part in theory and in the lab !